MOHAMAD HASHEM DANKAR

Architect · VR & digital twin specialist

An architect who builds cities twice.

BASED

ABU DHABI
UNITED ARAB EMIRATES

CURRENTLY

VR CONTENT CREATOR
UAE MINISTRY OF INTERIOR

PREVIOUSLY

FIBREX CONSTRUCTION GROUP
LDRS CHEHAB & PARTNERS

CONTACT

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STATEMENT

ONE CITY GETS BUILT IN CONCRETE. THE OTHER GETS BUILT IN *an engine.* I DO BOTH.

Six years across design practice, museum fabrication and government simulation. On Saadiyat Island I took three museums from design surface to cast panel — production 3D, GRC sheets, steel shop drawings, then the factory floor. A curved monument has to be cast flat, tracked, trucked and re-assembled true, and that is a geometry problem before it is anything else.

In Abu Dhabi I build the **UAE's national digital twin**: seven emirates modelled, textured, geolocated and running in real time in Unreal Engine, with an Arabic-first operational layer sitting on top of them. Not a fly-through — a working picture of a country, accurate enough to be simulated against.

Before the Gulf: regeneration for UNDP in Tripoli, heritage diagnosis on an 1878 monastery, competitions for Erbil, and a virtual-reality complex argued from a survey before a line was drawn. The through-line is the same at every scale — **build it accurately enough that it can be tested, not just admired.**

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PROJECTS

07
EMIRATES IN ENGINE

03
SAADIYAT MUSEUMS

117
BUILDINGS · REEM ISLAND

6+
YEARS IN PRACTICE

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REEM ISLAND IN ENGINE · ONE HUNDRED AND SEVENTEEN BUILDINGS, MODELLED, TEXTURED AND PINNED TO THEIR REAL LATITUDE

PROFILE & CREDENTIALS

Mohamad Hashem Dankar — architect (BAU, RIBA Part I), based in Abu Dhabi. **Six years across design practice, museum fabrication and government simulation**, working in English and Arabic, at every scale from a GRC bracket to a national twin.

CURRENT	VR Content Creator — UAE Ministry of Interior (Tahaluf Al Emarat), Abu Dhabi
PREVIOUSLY	Fibrex Construction Group, Saadiyat museums · LDRS Chehab & Partners, Beirut–Tripoli · independent practice
EDUCATION	B.Arch, Beirut Arab University — validated at RIBA Part 1 level
CREDENTIALS	Autodesk Certified User, 3ds Max & AutoCAD · IC3 GS5 · Sapienza Rome parametric programme
LANGUAGES	Arabic · English · French
CONTACT	hashemx98@gmail.com · +971 50 970 2011

TOOLCHAIN

- Unreal Engine
- 3ds Max
- Datasmith
- Substance
- Rhino + Grasshopper
- Revit
- AutoCAD
- Corona / V-Ray
- Twinmotion
- Unity
- Ladybug / Honeybee
- Photoshop
- GIS workflows

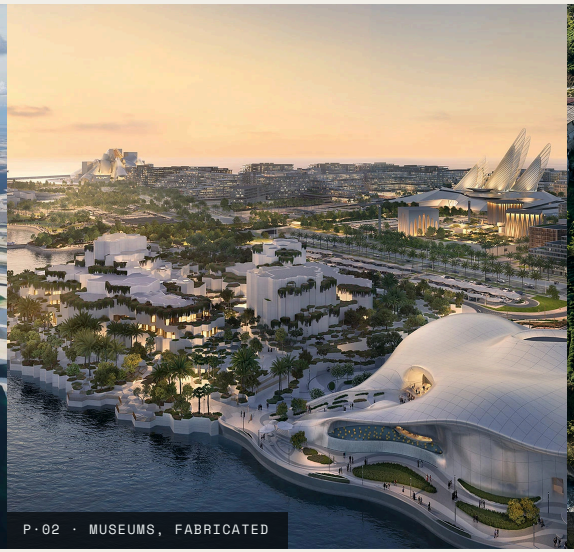
03
SAADIYAT MUSEUMS

07
EMIRATES IN ENGINE

117
BUILDINGS · REEM ISLAND

6+
YEARS IN PRACTICE

P·01 · UAE DIGITAL TWIN


P·02 · MUSEUMS, FABRICATED

P·03 · FREELANCE PRACTICE

P·04 · PRACTICE YEARS

CHAPTER ONE · FLAGSHIP

01

UAE DIGITAL TWIN

A national-scale digital twin of the United Arab Emirates, running in real time in **Unreal Engine**. I build the cities themselves — modelled in 3ds Max, textured in Substance, delivered through Datasmith, geolocated against GIS.

CLIENT	EMPLOYER	SCOPE	STATUS
UAE MINISTRY OF INTERIOR	TAHALUF AL EMARAT	SEVEN EMIRATES · NATIONAL SCALE	IN ACTIVE DEVELOPMENT

1.1

THE PIPELINE

SATELLITE → GIS → 3DS MAX → DATASMITH → UNREAL

Every city in the twin goes through the same disciplined route, built for **speed, consistency and real-time performance** at urban scale. When a landmark matters it is rebuilt properly: grey massing to fully textured, night-lit asset.

01

SATELLITE / GIS

Imagery and footprints, georeferenced to the real coordinate frame

02

2D EXTRACTION

Clean building outlines, district by district

03

3D LOW-POLY

LOD 100–200 massing, modelled in 3ds Max

04

UV + TEXTURE

Box mapping · custom and emissive maps in Substance

05

DATASMITH

Pivots, naming, tags and collision, ready for engine

06

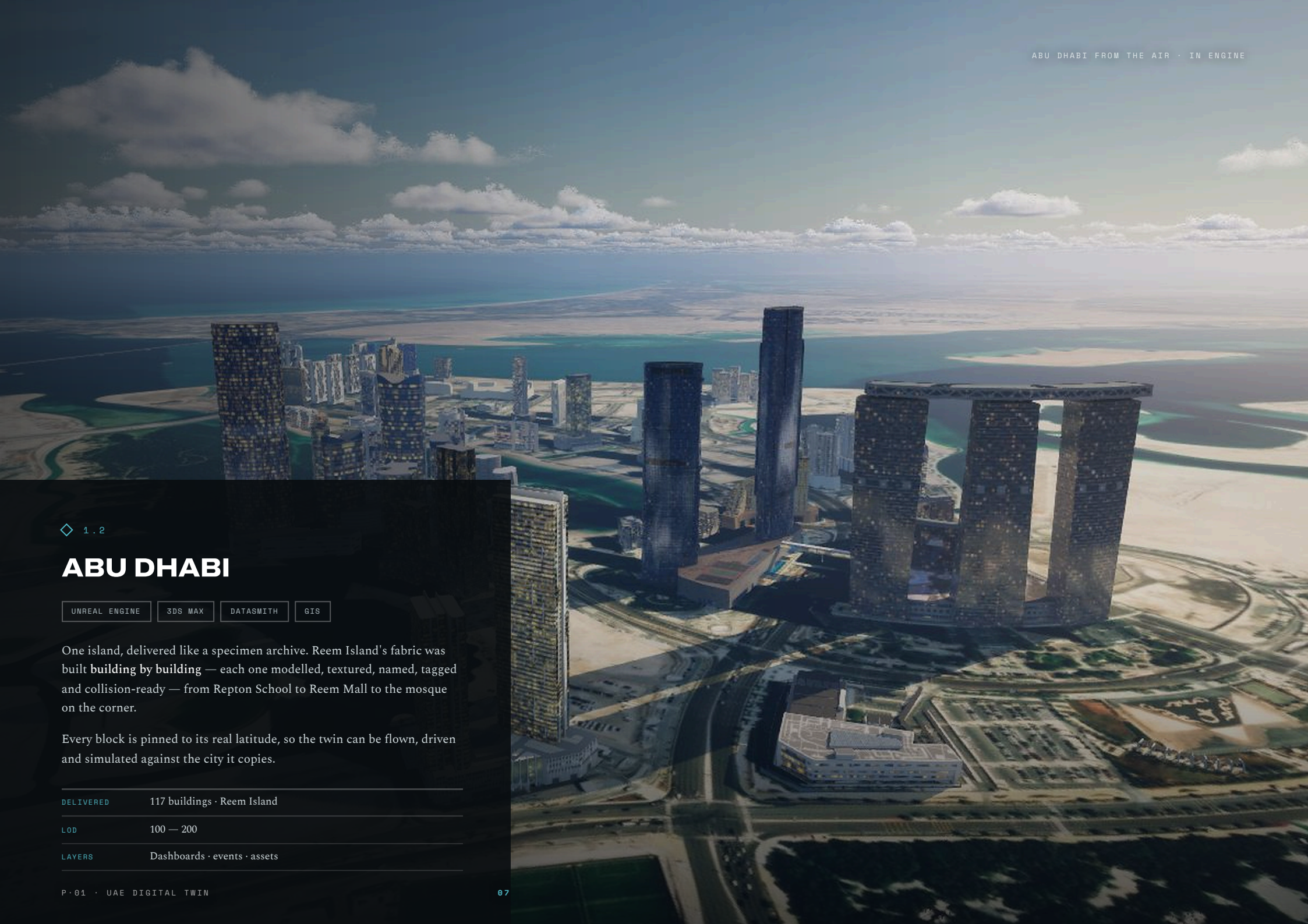
UNREAL ENGINE

Geolocation · blueprints · events · the dashboard layer

THE SAME BUILDING, TWICE — GREY MASSING TO NIGHT-LIT ASSET

SIGMA & OMEGA TOWERS · MASSING → TEXTURED NIGHT ASSET

MANGROVE COMPLEX · EXPLODED TEXTURE SET → ASSEMBLED



ABU DHABI FROM THE AIR · IN ENGINE

1.2

ABU DHABI

UNREAL ENGINE | 3DS MAX | DATASMITH | GIS

One island, delivered like a specimen archive. Reem Island's fabric was built **building by building** — each one modelled, textured, named, tagged and collision-ready — from Repton School to Reem Mall to the mosque on the corner.

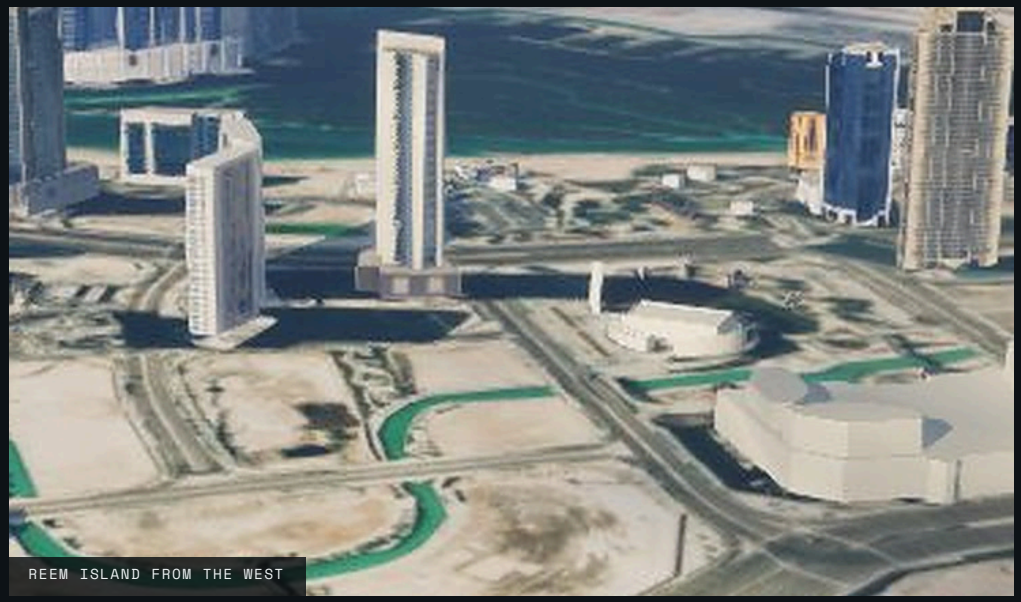
Every block is pinned to its real latitude, so the twin can be flown, driven and simulated against the city it copies.

DELIVERED	117 buildings · Reem Island
LOD	100 — 200
LAYERS	Dashboards · events · assets

1.2

THE ISLAND, BLOCK BY BLOCK

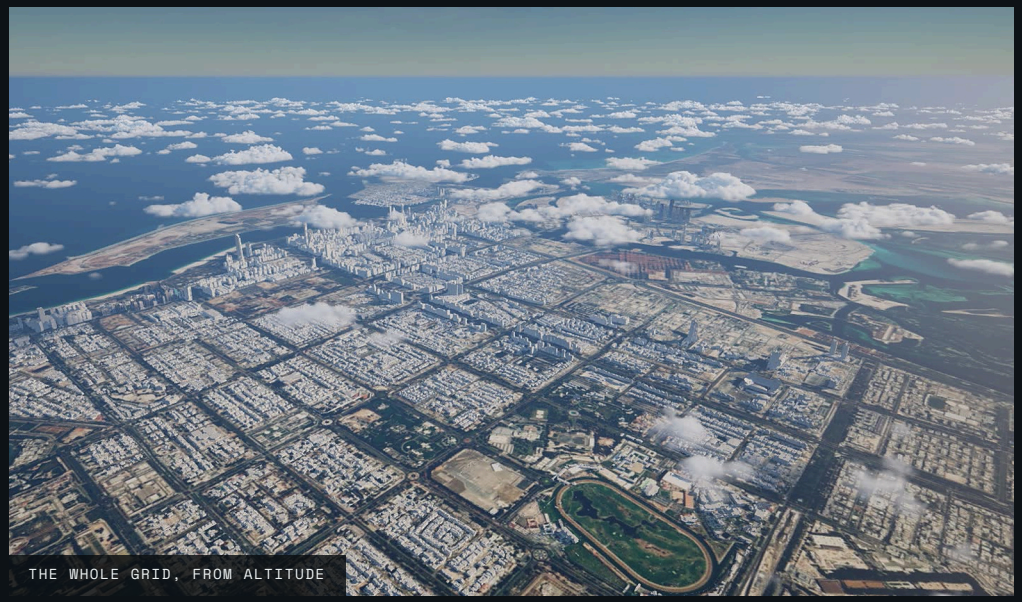
The same disciplined route runs under every city in the twin: footprint extraction, low-poly massing, UV and texture, Datasmith delivery, then geolocation and blueprints in engine. What comes out is not a fly-through — it is a place with coordinates.



REEM ISLAND FROM THE WEST

STREET LEVEL · THE REEM MALL FRONTAGE

AL MARYAH ISLAND

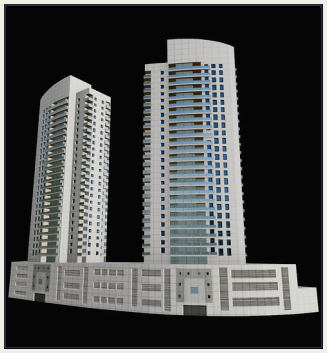


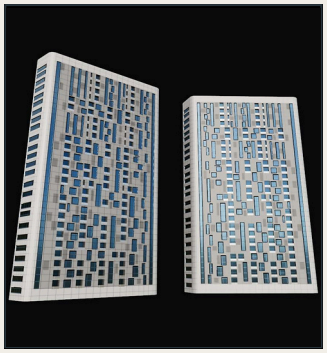
THE WHOLE GRID, FROM ALTITUDE

THE ASSET VAULT

All eighteen named Reem Island buildings, pulled straight from the delivery archive. Each one is a real engine asset — materials, night lights, collision, pivot and tag — not a massing block standing in for one.

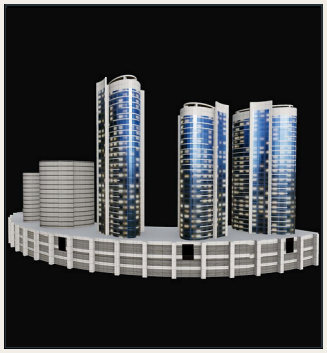
AL WIFAQ TOWER


AMAYA TOWERS


SHAMS MEERA TOWERS

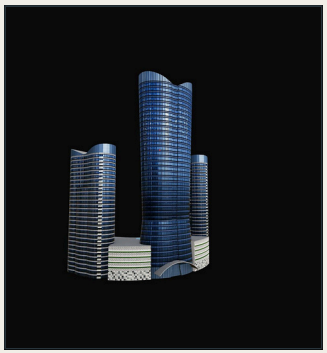
THE KITE RESIDENCES

MARINA RISE TOWER

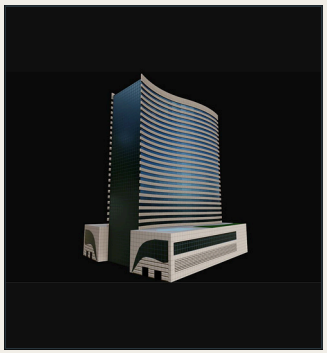

HYDRA TOWERS

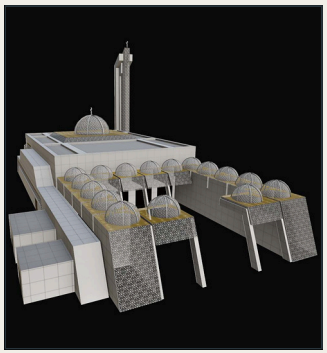
MANGROVE PLACE

REEM HOSPITAL

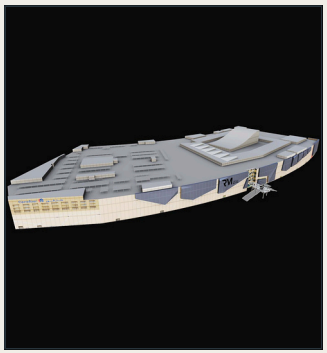

SIGMA OMEGA TOWERS

AL REEM BAY TOWERS


DAMAC MARINA BAY

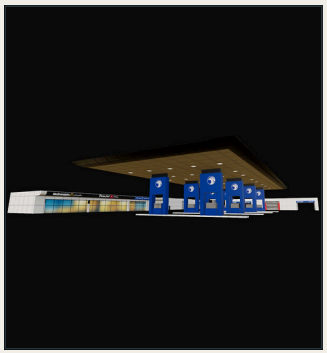

REEM MOSQUE

PARAGON BAY MALL

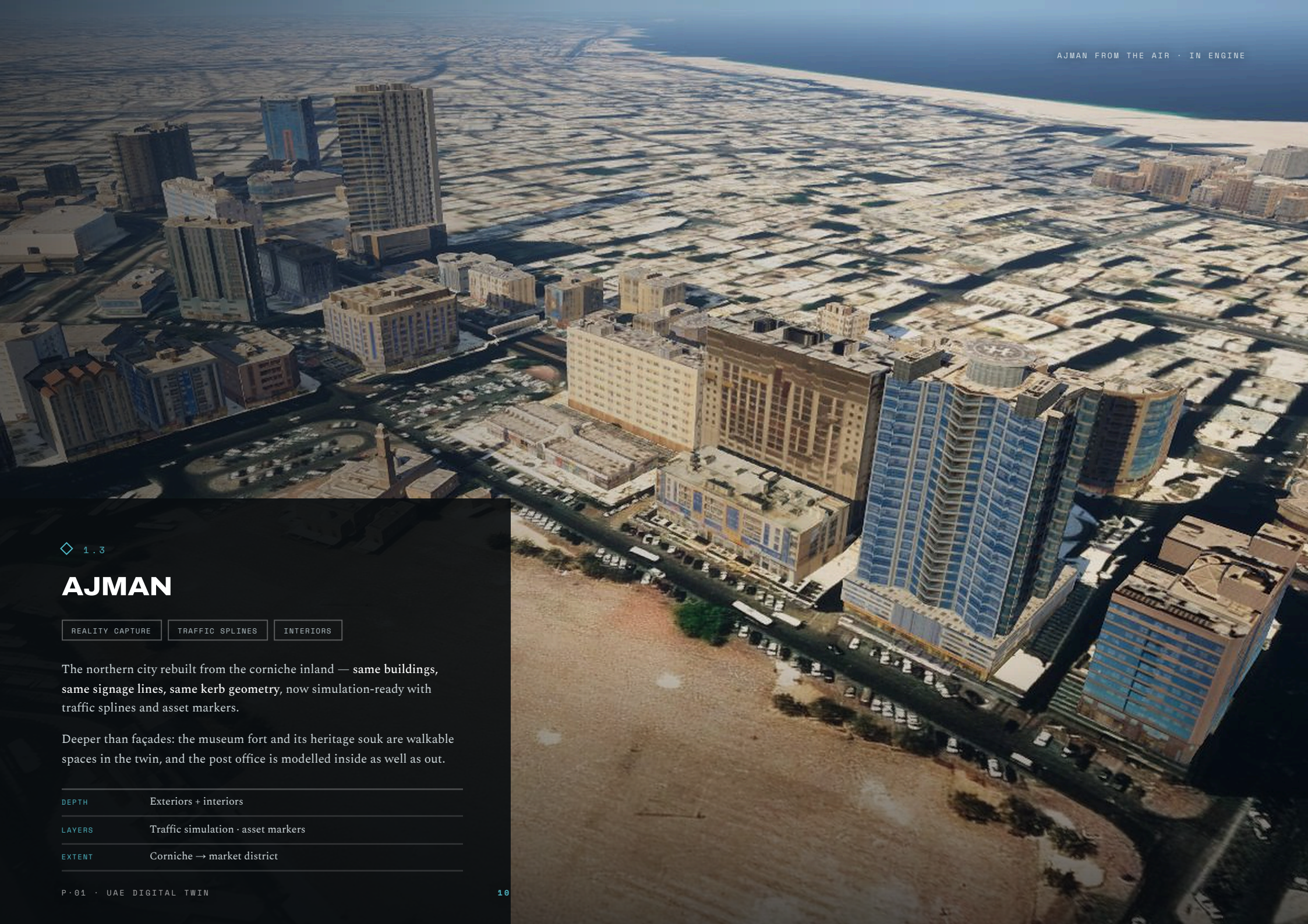

REEM MALL

SORBONNE UNIVERSITY

Y TOWER


ADNOC STATION

REPTON SCHOOL · ROSE CAMPUS



AJMAN FROM THE AIR · IN ENGINE

◇ 1.3

AJMAN

REALITY CAPTURE | TRAFFIC SPLINES | INTERIORS

The northern city rebuilt from the corniche inland — **same buildings, same signage lines, same kerb geometry**, now simulation-ready with traffic splines and asset markers.

Deeper than façades: the museum fort and its heritage souk are walkable spaces in the twin, and the post office is modelled inside as well as out.

DEPTH	Exteriors + interiors
LAYERS	Traffic simulation · asset markers
EXTENT	Corniche → market district

ONE SEAM APART

AJMAN, UAE · THE PHOTOGRAPHED STREET AND THE SAME STREET IN ENGINE, SPLIT DOWN THE MIDDLE

NOTHING APPROXIMATED

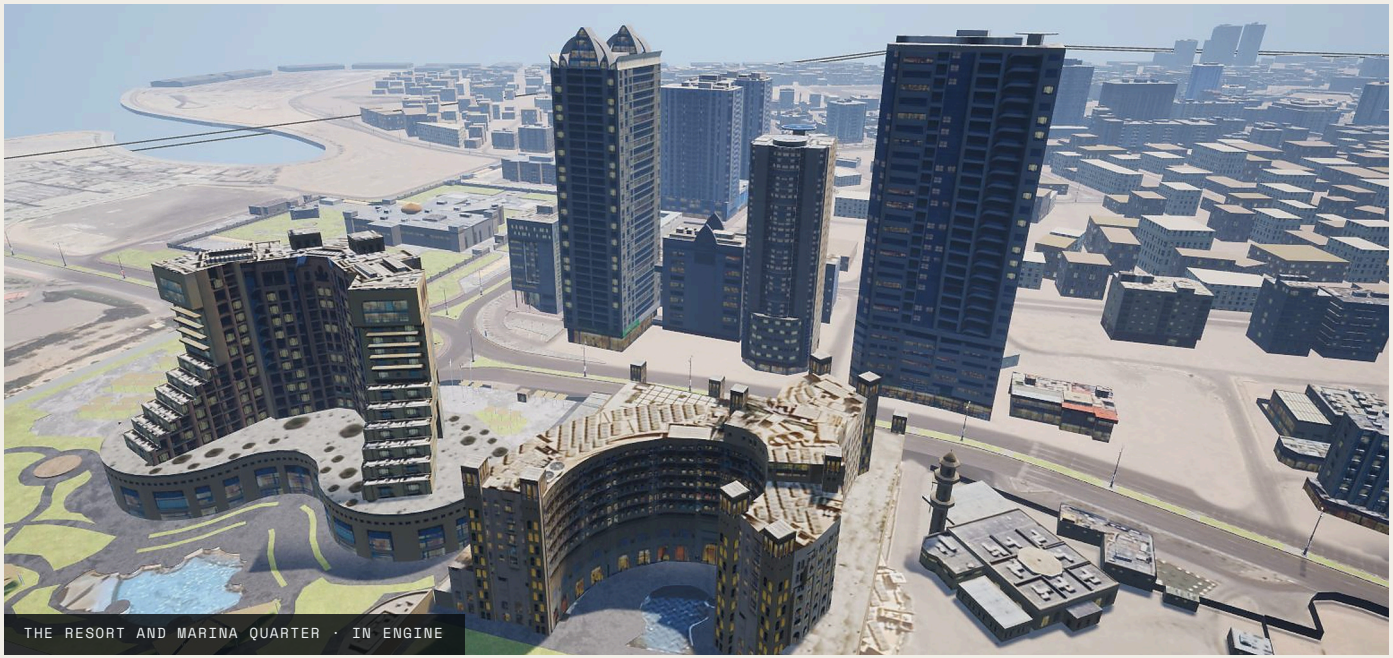
The proof sits in a single frame. On one side of that seam is the real Ajman corner; on the other, the same corner inside the engine. The tower spacing, the shop signage, the kerb radius and the palm line all carry through.

That is the whole test of a twin. A fly-through only has to look right from one camera. A simulation has to hold up when a vehicle turns the corner, when a crowd is routed down the pavement, when an incident is placed on the roundabout and the model has to answer for it.

So the city is built to be walked into, not admired from altitude — dhow monument, market blocks, marina quarter, each one carrying its real position.

METHOD	Photo survey · georeferenced
OUTPUT	Simulation-ready street
DEPTH	Exterior and interior

THE DHOW MONUMENT · REBUILT WHERE IT STANDS

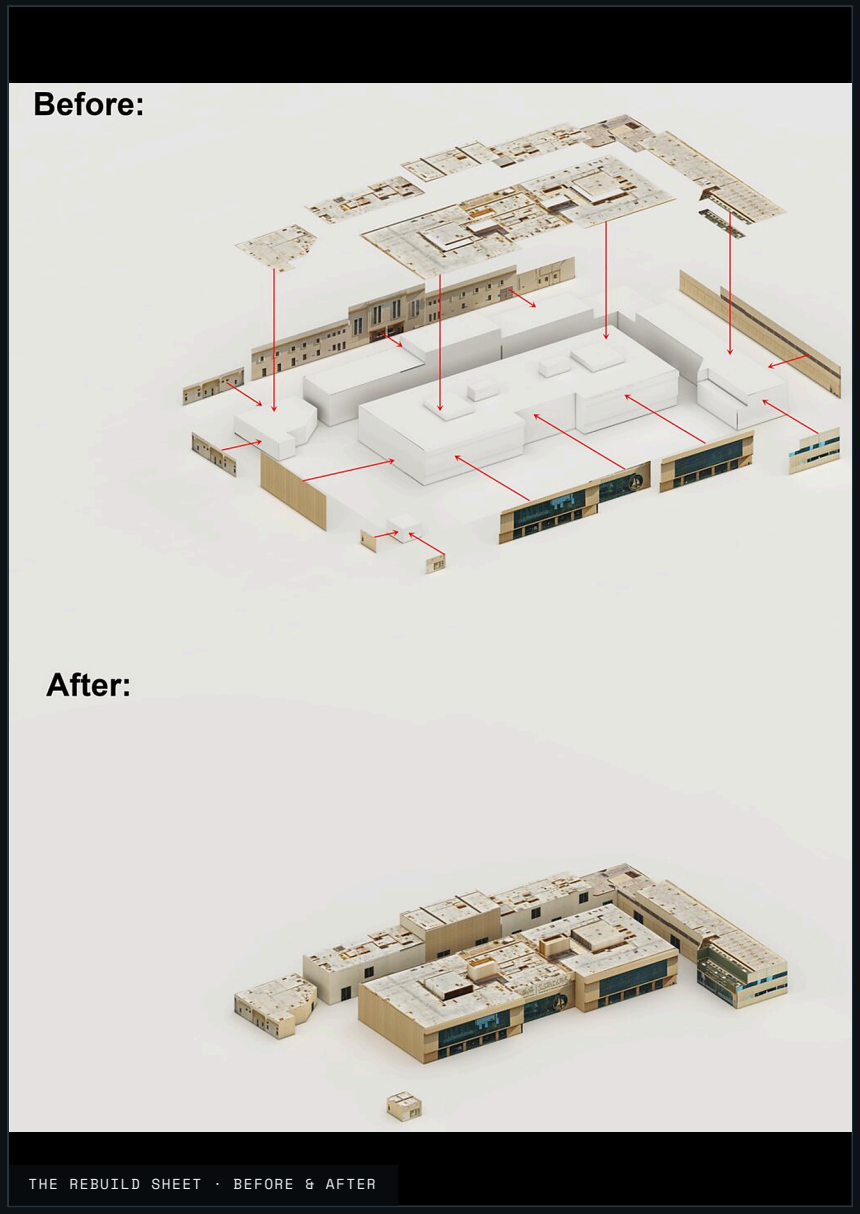


THE RESORT AND MARINA QUARTER · IN ENGINE

1.3

ONE BUILDING, THREE WAYS

A municipality building on a street in Ajman, taken through the whole route in one sheet: the photograph, the massing, the textured asset. Put the three states beside each other and **any drift shows immediately** — a window band in the wrong place, a parapet a storey too high, a colour that never stood on the street.



THE REBUILD SHEET · BEFORE & AFTER

REALITY · PHOTOGRAPHED ON THE STREET

THE TWIN · STANDING IN ENGINE

1.3

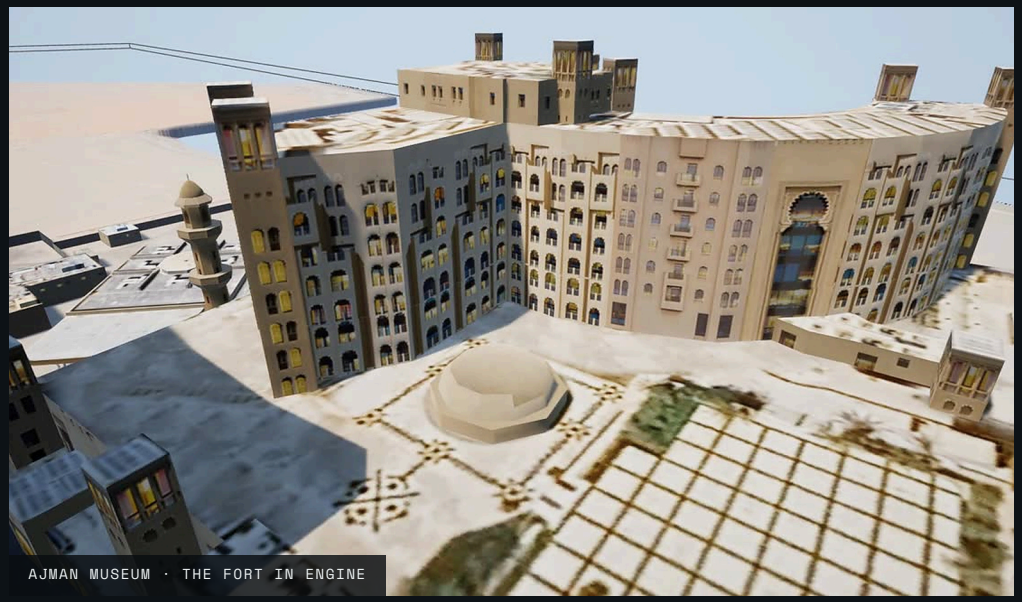
INSIDE THE TWIN

Where it matters, the twin goes indoors. The post office is modelled inside as well as out, and the museum fort and its heritage souk are walkable rooms rather than façades with nothing behind them.

AJMAN POST OFFICE · TRAFFIC SIMULATION LAYER

…AND INSIDE IT — INTERIORS ARE IN THE TWIN TOO

THE HERITAGE SOUK · WALKED IN ENGINE



AJMAN MUSEUM · THE FORT IN ENGINE

1.4

RAS AL KHAIMAH

GIS FOOTPRINTS | COASTAL STRIP | LAGOON

The northern emirate, rebuilt from the mountains to the lagoon. **The corniche, the market district and the coastal strip** — every block extracted from GIS footprints, massed, textured and pinned to its real latitude.

EXTENT	Creek · lagoon · corniche
SOURCE	GIS footprints + imagery

THE HIGH-ALTITUDE PASS

THE CORNICHE · WATERFRONT ROW

THE LAGOON, AT WATER LEVEL

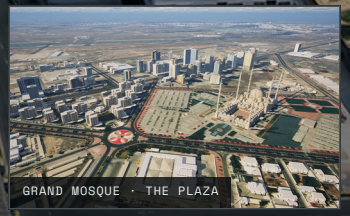
1.5

FUJAIRAH

HAJAR FOOTHILLS | GRAND MOSQUE AXIS | PORT

The east-coast emirate, from the Hajar foothills to the port. **The Grand Mosque axis, the Hamad Bin Abdullah boulevard and the oil terminal** — modelled, textured and running in the same national world.

AXIS	Grand Mosque → city centre
INDUSTRY	Oil terminal · tank farm


GRAND MOSQUE · THE PLAZA

CITY CENTRE · IN ENGINE

THE OIL TERMINAL

1.6

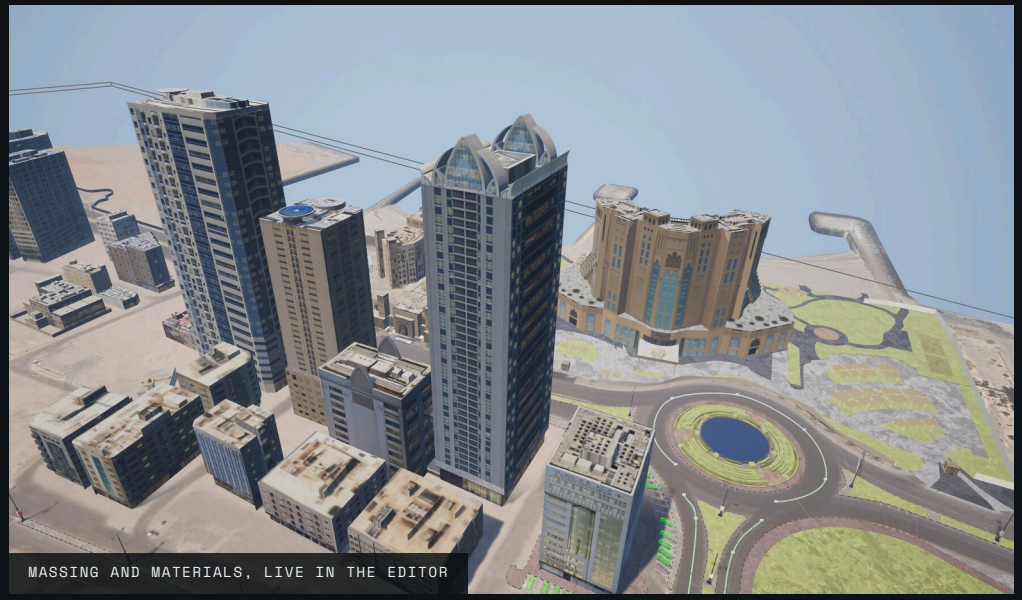
THE LAYER ON TOP

The cities are the substrate. On top of them sits an **Arabic-first readiness dashboard**: the twin is not a fly-through but a working operational picture, with live data pinned into real geography.

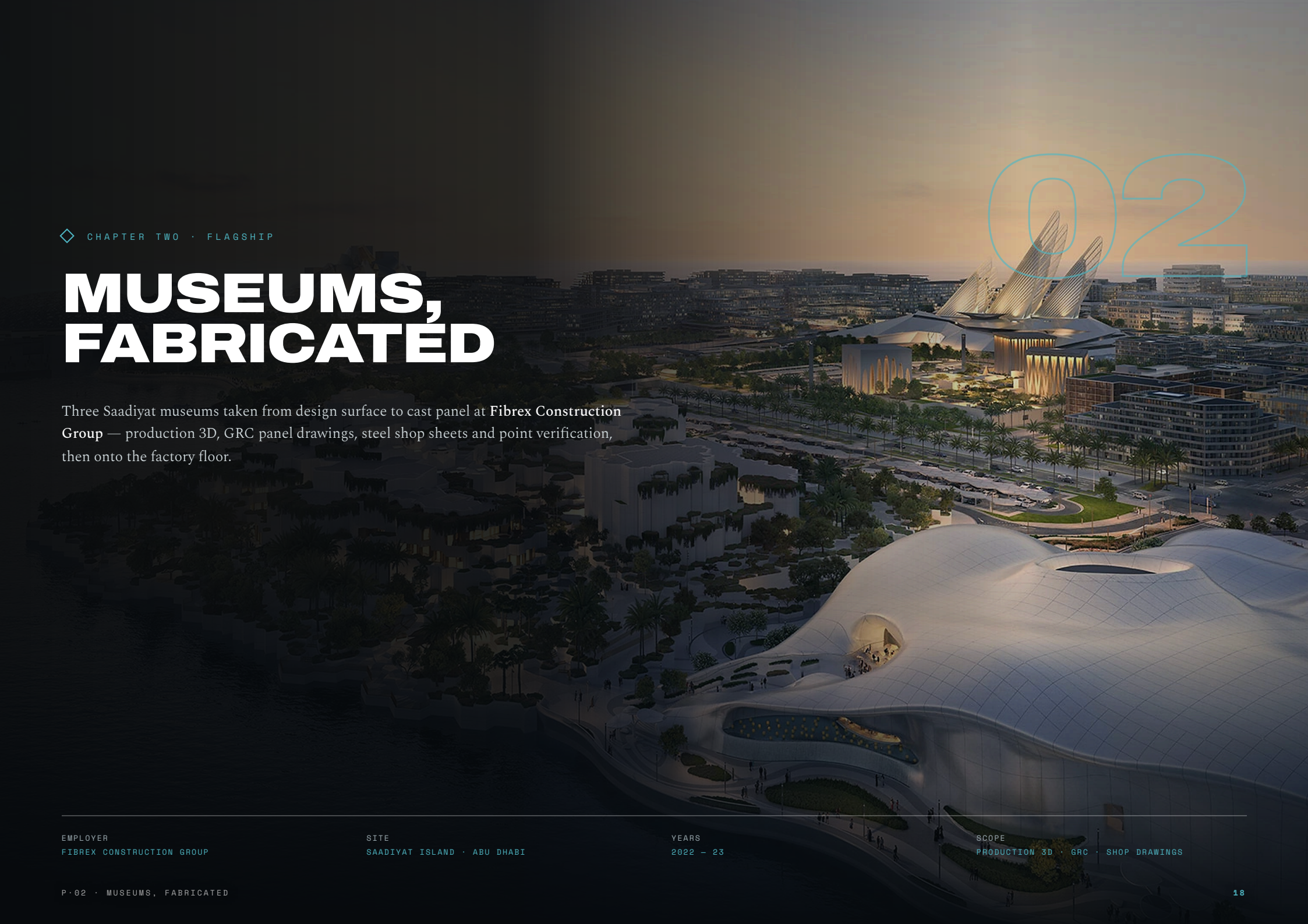
READINESS DASHBOARD · ABU DHABI

READINESS DASHBOARD · THE COAST

READINESS DASHBOARD · RAS AL KHAIMAH



MASSING AND MATERIALS, LIVE IN THE EDITOR



CHAPTER TWO · FLAGSHIP

02

MUSEUMS, FABRICATED

Three Saadiyat museums taken from design surface to cast panel at **Fibrex Construction Group** — production 3D, GRC panel drawings, steel shop sheets and point verification, then onto the factory floor.

EMPLOYER	SITE	YEARS	SCOPE
FIBREX CONSTRUCTION GROUP	SAADIYAT ISLAND · ABU DHABI	2022 — 23	PRODUCTION 3D · GRC · SHOP DRAWINGS

◇ 2.1

ZAYED NATIONAL MUSEUM

PRODUCTION 3D | STEEL WINGS | GRC | SHOP DRAWINGS

The museum's falcon-wing towers pass through the factory as thousands of unique elements. I produced the **production 3D models, panel breakdowns and steel shop-drawing sheets** — the geometry that lets a curved monument be cast flat, tracked, trucked and re-assembled on site.

The museum opened on Saadiyat in 2025.

ARCHITECT	Foster + Partners
EMPLOYER	Fibrex Construction Group
STATUS	Built · open

2.1

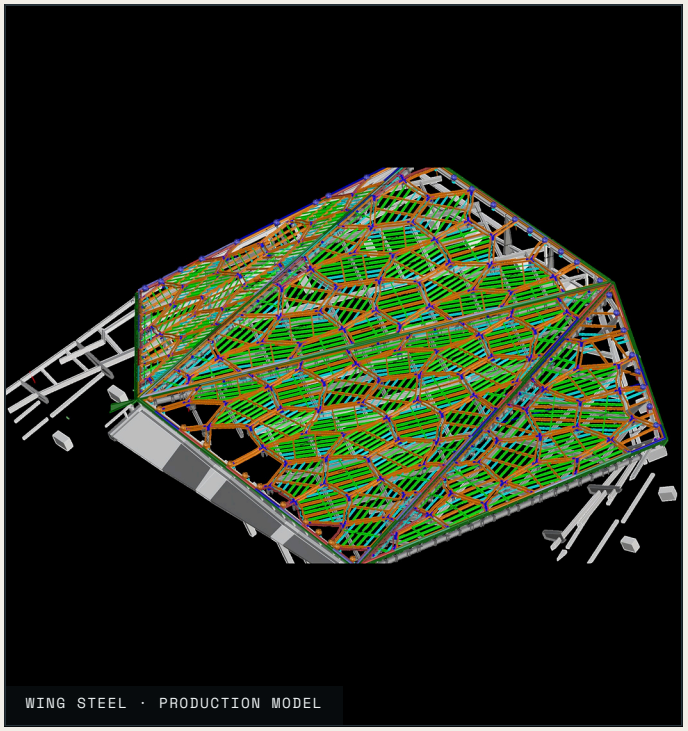
THE GEOMETRY BEHIND THE WING

A wing is not drawn once. It is broken into steel ribs, panel families and fixing points, each carrying its own sheet, its own number and its own tolerance.

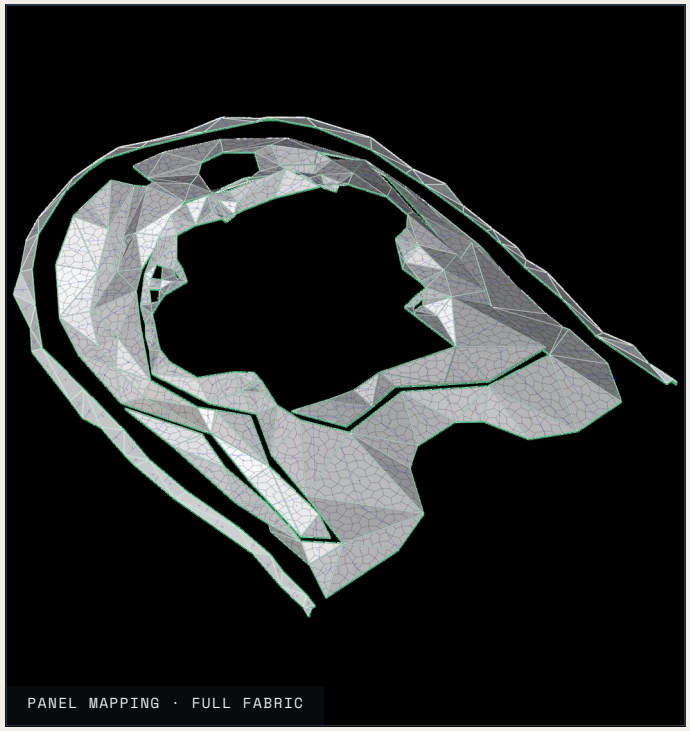
The wing steel is modelled as it will be welded, not as it is drawn in the design model. Panels are mapped across the whole fabric so that no two neighbours share a number, and every bracket is detailed to the bolt.

Get one sheet wrong and a piece arrives on Saadiyat that fits nothing. The discipline is the deliverable.

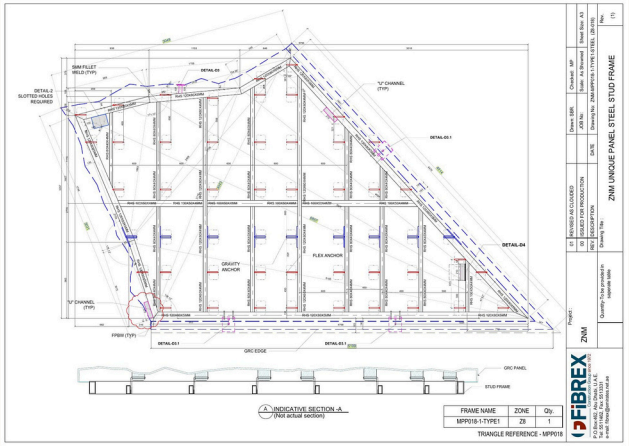
DELIVERABLE	Production 3D · panel breakdowns
SHEETS	Steel shop drawings, per element
SCALE	Thousands of unique elements



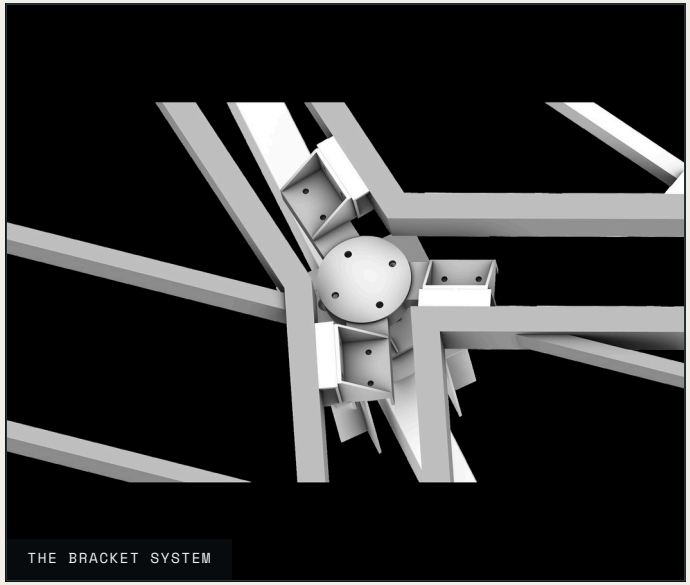
WING STEEL · PRODUCTION MODEL



PANEL MAPPING · FULL FABRIC



STEEL SHOP SHEET · Z8-018

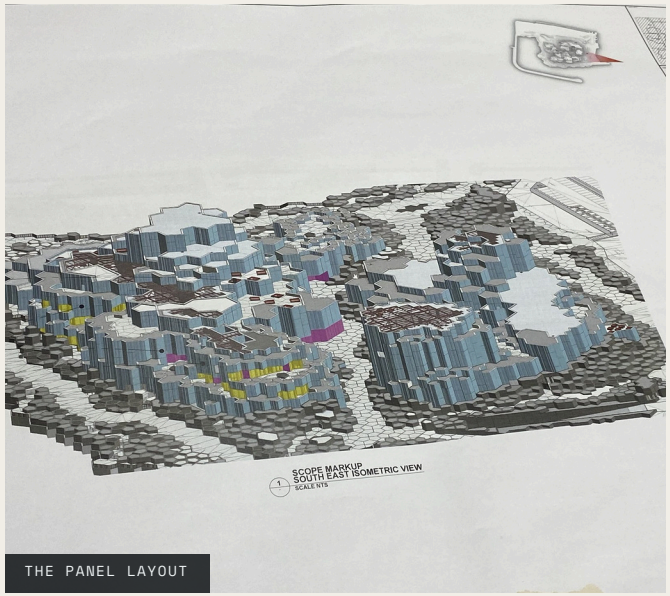


THE BRACKET SYSTEM

2.1

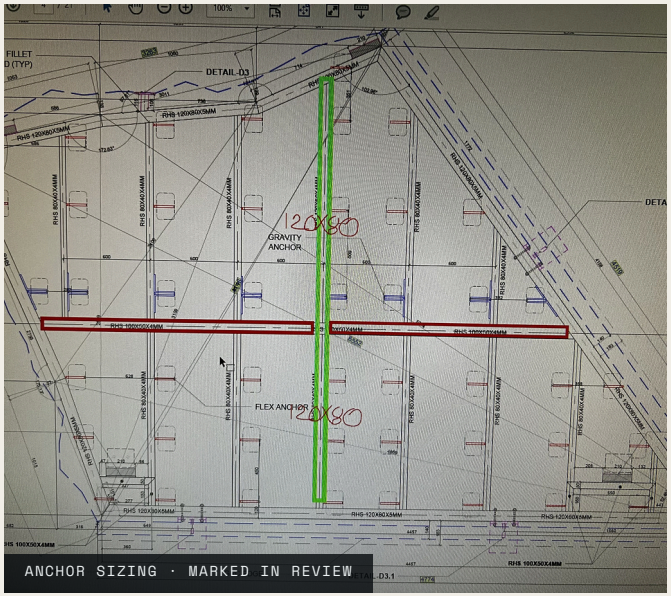
DOWN THE LINE

From the numbered layout to the saw, to the anchor, to the finished faceted stone — the same panel followed all the way through the works.

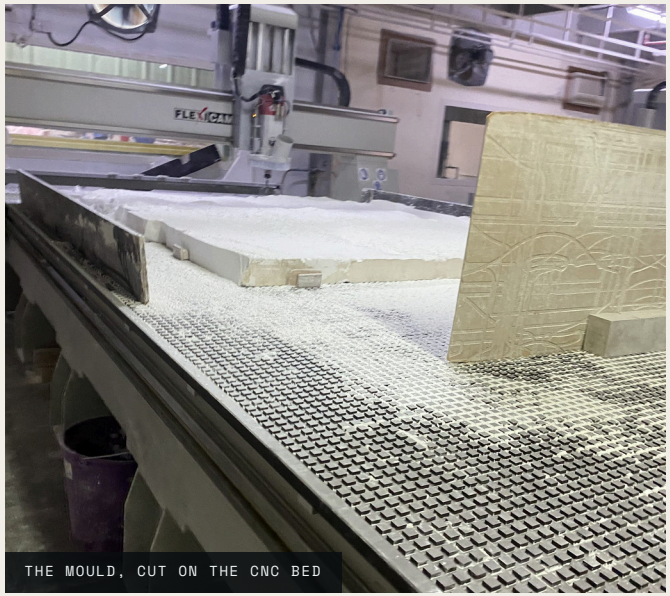


THE PANEL LAYOUT

RACKED IN ROWS



ANCHOR SIZING · MARKED IN REVIEW



THE MOULD, CUT ON THE CNC BED

A FINISHED FACETED PANEL

WING ASSEMBLIES, READY FOR DELIVERY

2.2

GUGGENHEIM ABU DHABI

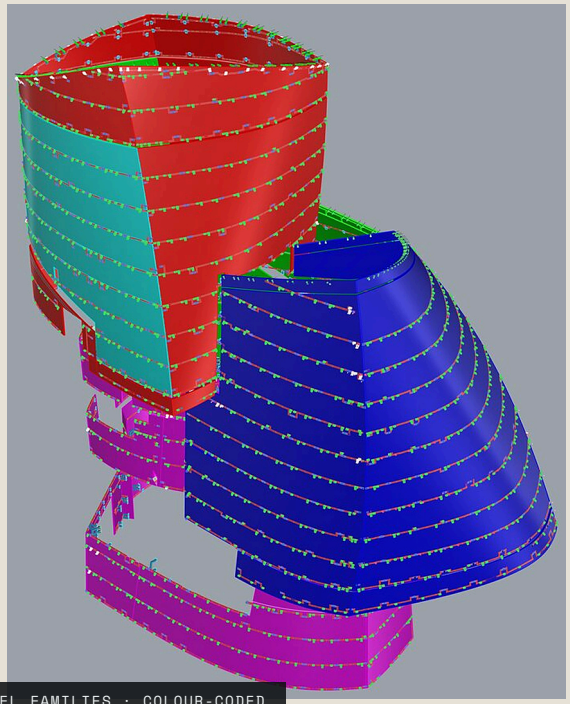
For Gehry's colliding cones I prepared **panel production models and point-cloud verification geometry** — the survey-grade 3D that proves each cast GRC panel matches the design surface before it leaves the yard.

A GRC panel is cast against a mould, not against the building. The verification geometry closes that gap: every fixing point is carried as a coordinate, checked against the cast piece in the yard, and signed off before it is loaded.

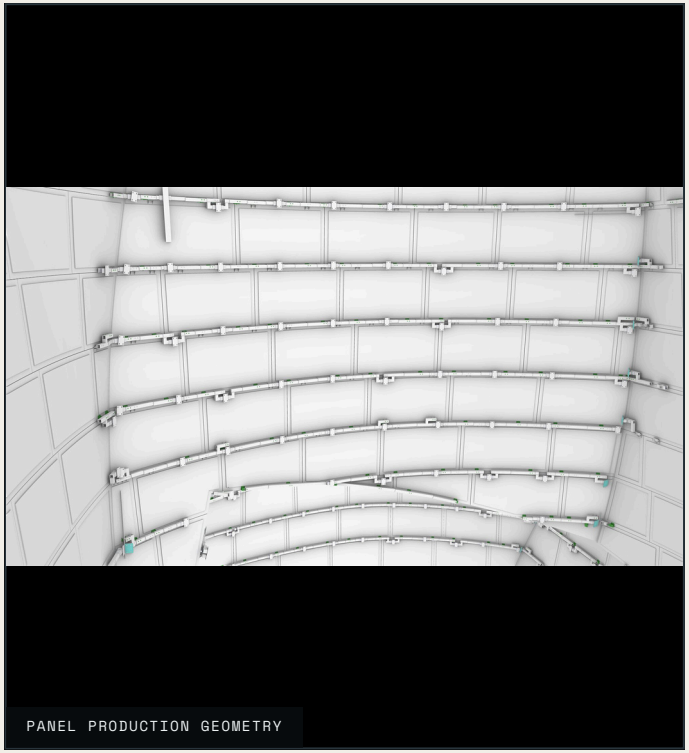
By the time a panel reaches the crane, its position on the design surface has been proved twice — once in the production model, once on the survey table.

ARCHITECT	Gehry Partners
SCOPE	GRC panel production · verification
SITE	Saadiyat Island · Abu Dhabi

THE COLLIDING CONES · PRESENTATION MODEL



PANEL FAMILIES · COLOUR-CODED

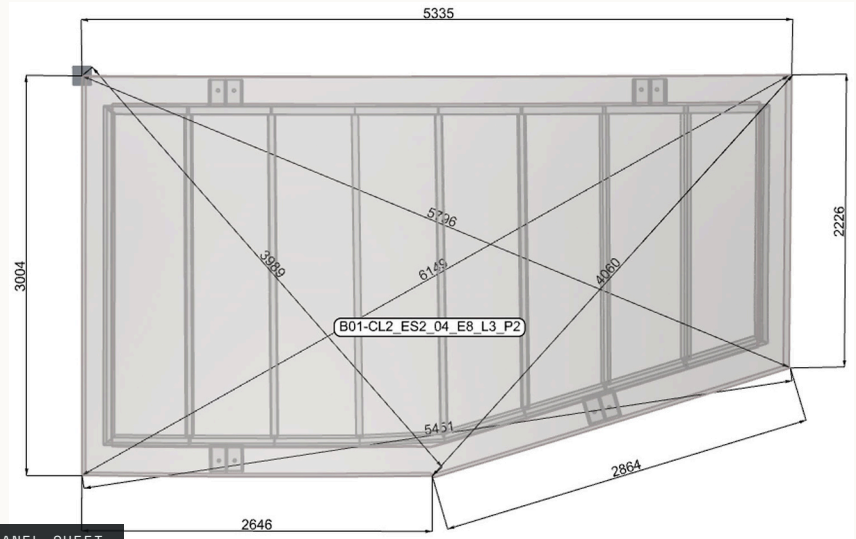


PANEL PRODUCTION GEOMETRY

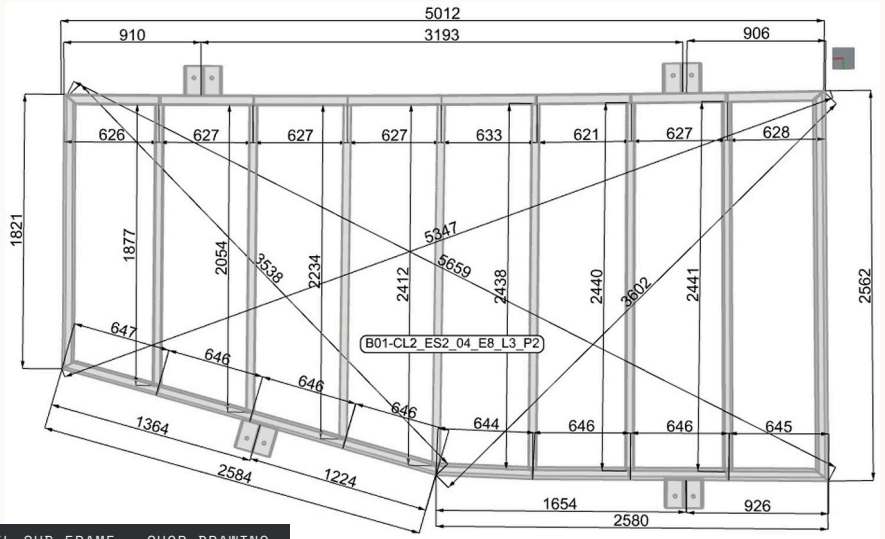
2.2

THE PAPERWORK BEHIND ONE PANEL

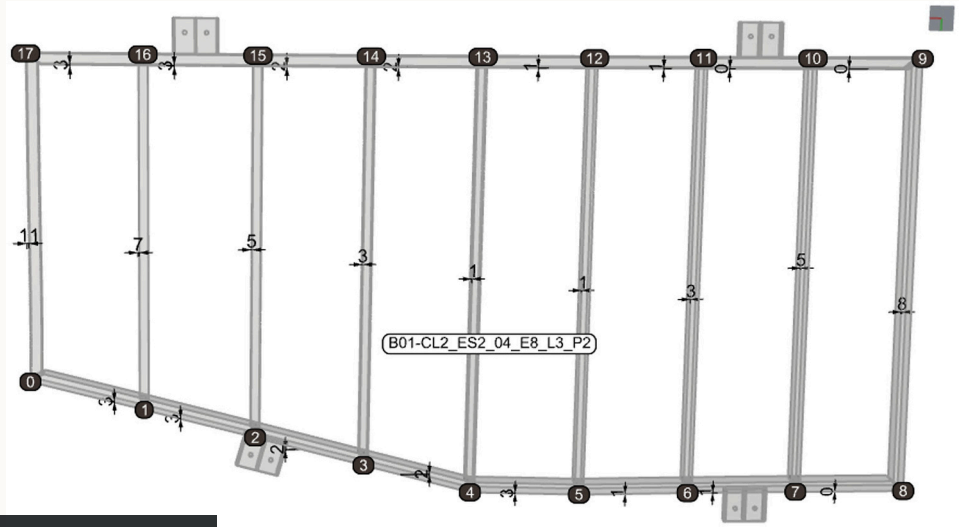
Behind a single cast piece sit the GRC sheet, its steel sub-frame, the fixing-point layout and the survey table that proves the panel lands exactly where the design surface says it should.



GRC PANEL SHEET



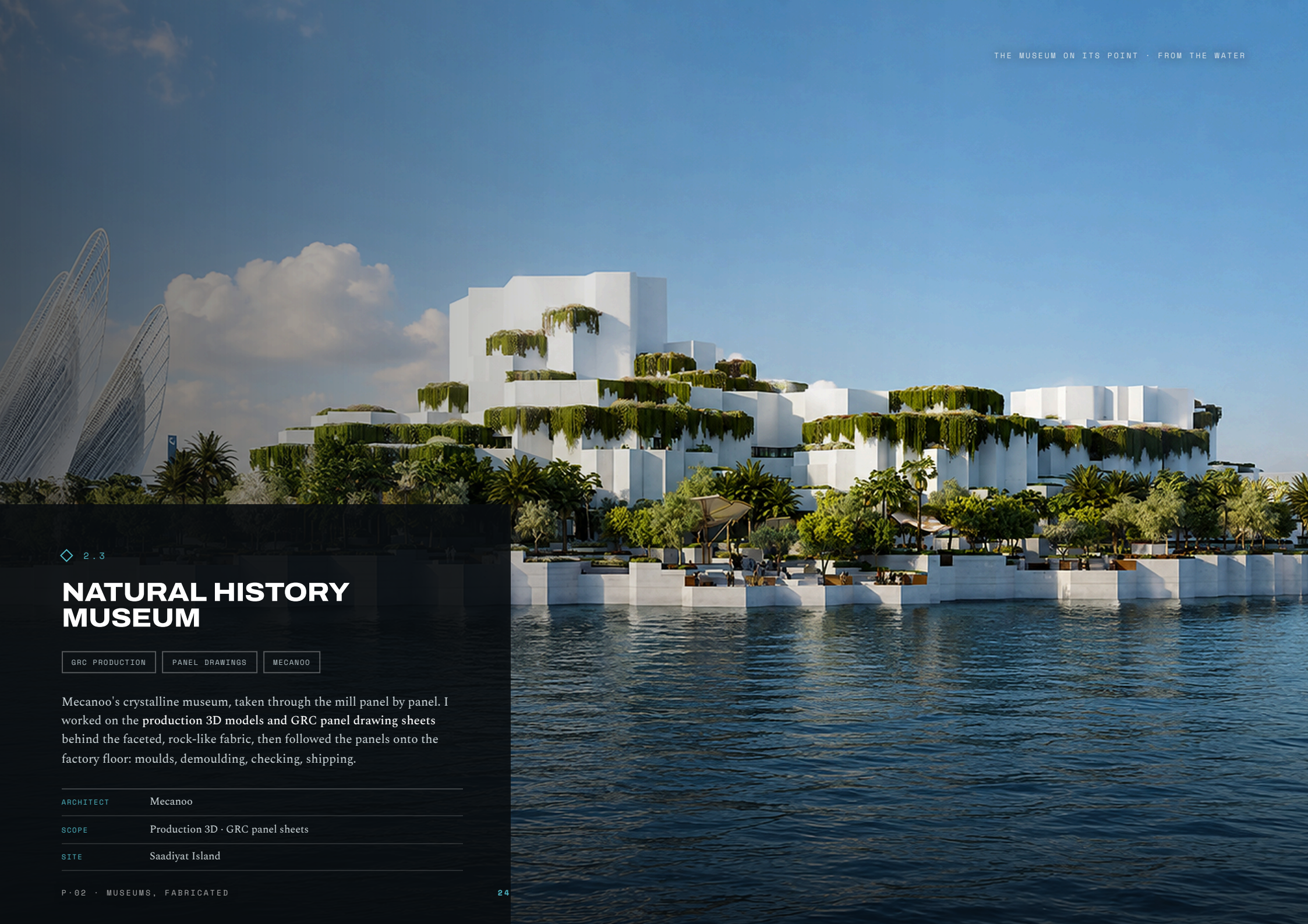
STEEL SUB-FRAME · SHOP DRAWING



FIXING-POINT LAYOUT

		X	Y	Z
	{0;0}	{0}	{0}	{0}
0	PT0	108	235	433
1	PT1	754	229	417
2	PT2	1400	225	413
3	PT3	2046	220	421
4	PT4	2692	220	440
5	PT5	3317	372	460
6	PT6	3937	551	491
7	PT7	4556	732	535
8	PT8	5172	915	590
9	PT9	4442	3368	452
10	PT10	3844	3188	389
11	PT11	3244	3011	338
12	PT12	2651	2833	293
13	PT13	2042	2661	270
14	PT14	1439	2488	254
15	PT15	836	2318	250
16	PT16	231	2151	259
17	PT17	-372	1985	282

SURVEY COORDINATE TABLE



THE MUSEUM ON ITS POINT · FROM THE WATER

2.3

NATURAL HISTORY MUSEUM

GRC PRODUCTION | PANEL DRAWINGS | MECANOO

Mecanoo's crystalline museum, taken through the mill panel by panel. I worked on the **production 3D models and GRC panel drawing sheets** behind the faceted, rock-like fabric, then followed the panels onto the factory floor: moulds, demoulding, checking, shipping.

ARCHITECT	Mecanoo
SCOPE	Production 3D · GRC panel sheets
SITE	Saadiyat Island

2.3

MOULD, DEMOULD, CHECK, SHIP

The faceted fabric is cast piece by piece. Every panel is drawn, moulded, demoulded, checked in the yard and shipped — and the photograph from the site is the same geometry, standing.

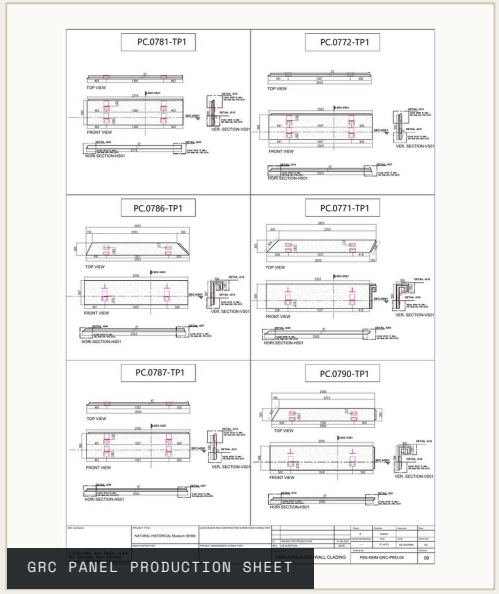
MOULDS IN PRODUCTION

DEMOULDED PANEL

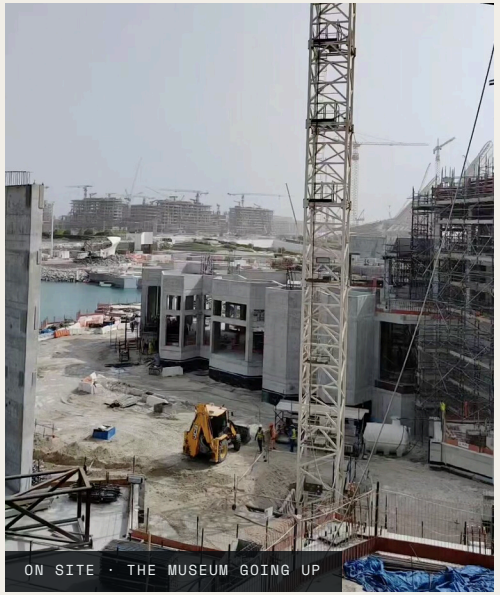
CASTING WORKS

CHECKED IN THE YARD

PRODUCTION MODEL OF THE FACETED GEOMETRY



GRC PANEL PRODUCTION SHEET



ON SITE · THE MUSEUM GOING UP

FROM THE WATER · THE STEPPED TERRACES

2.4

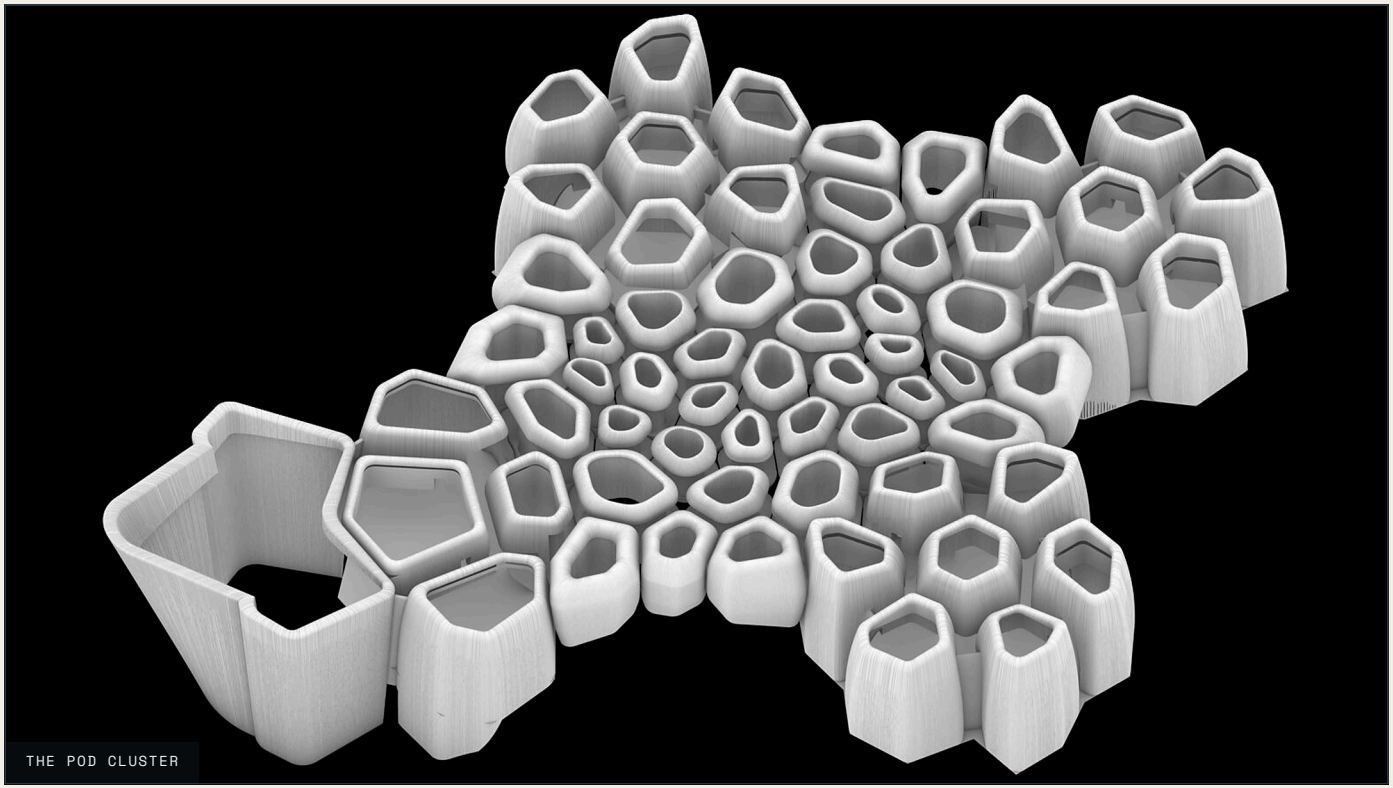
AMAALA RED SEA VILLAGE

On the Red Sea coast, a village of cellular pods. A **Grasshopper definition drives the cluster** from design surface to castable production units, so every pod can vary while staying buildable from the same family of moulds.

The definition is the deliverable as much as the render: change the shoreline and the village re-packs itself, and the pods stay castable.

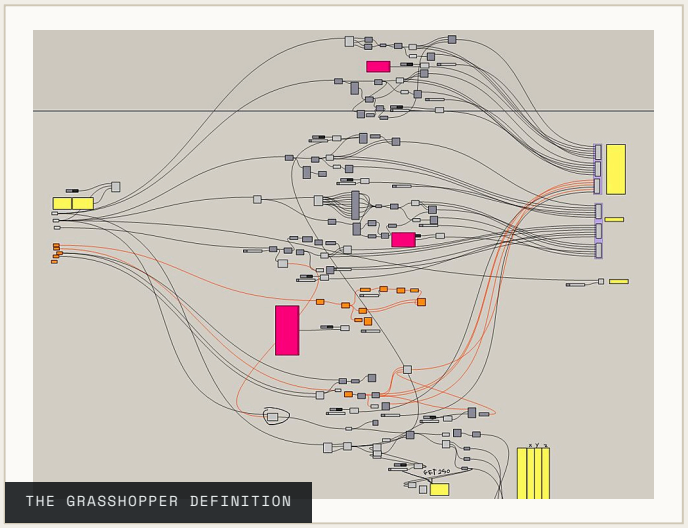
That is the point of driving it parametrically. A hand-drawn cluster of a hundred cells would take a week to revise and would quietly drift out of the mould family. A definition revises in an afternoon and cannot.

SITE	Amaala · Red Sea coast · KSA
TOOLING	Grasshopper · Rhino · 3ds Max
LOGIC	One mould family, many pods
OUTPUT	Castable production units

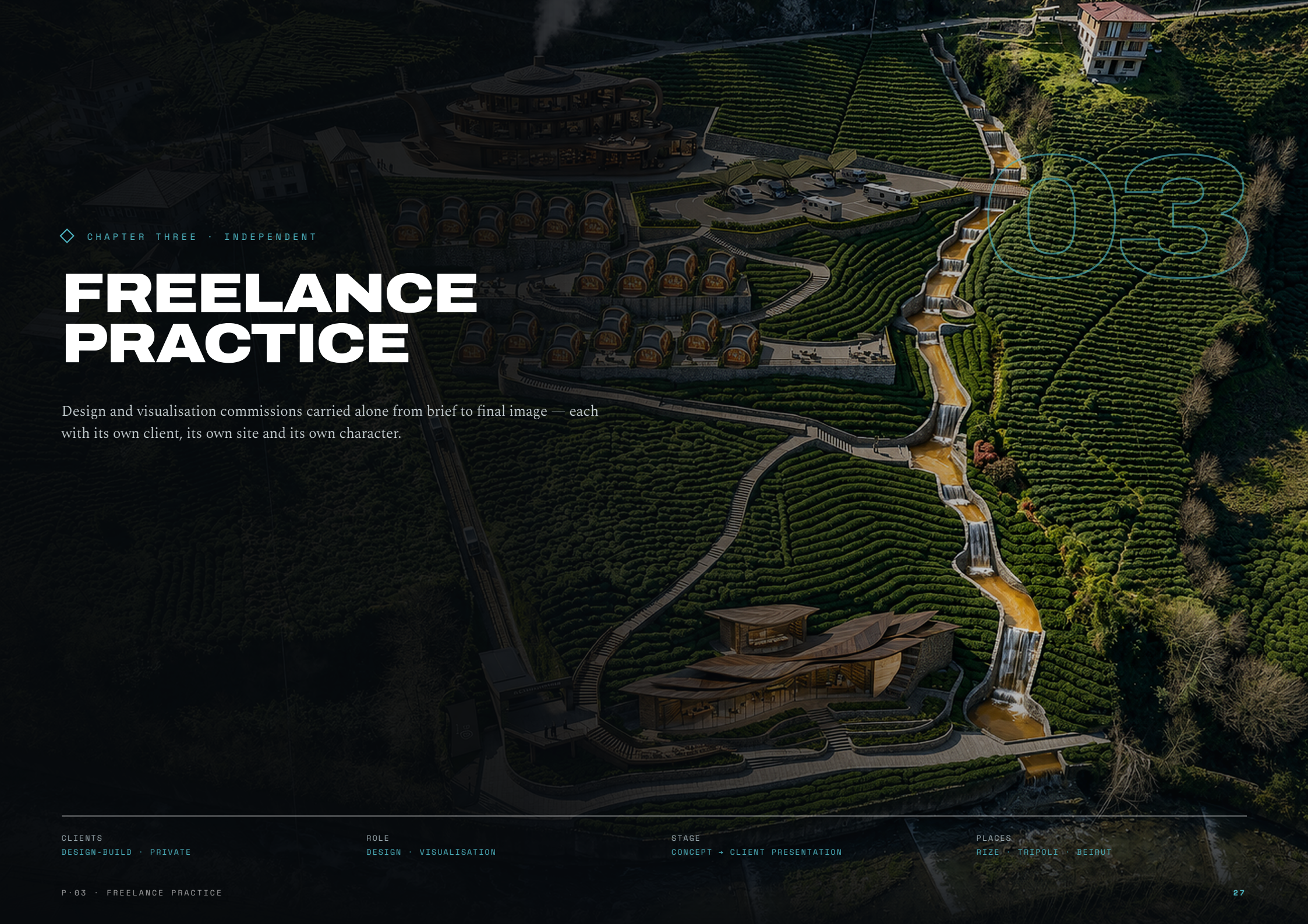


THE POD CLUSTER

SHORELINE AERIAL



THE GRASSHOPPER DEFINITION



CHAPTER THREE · INDEPENDENT

FREELANCE PRACTICE

03

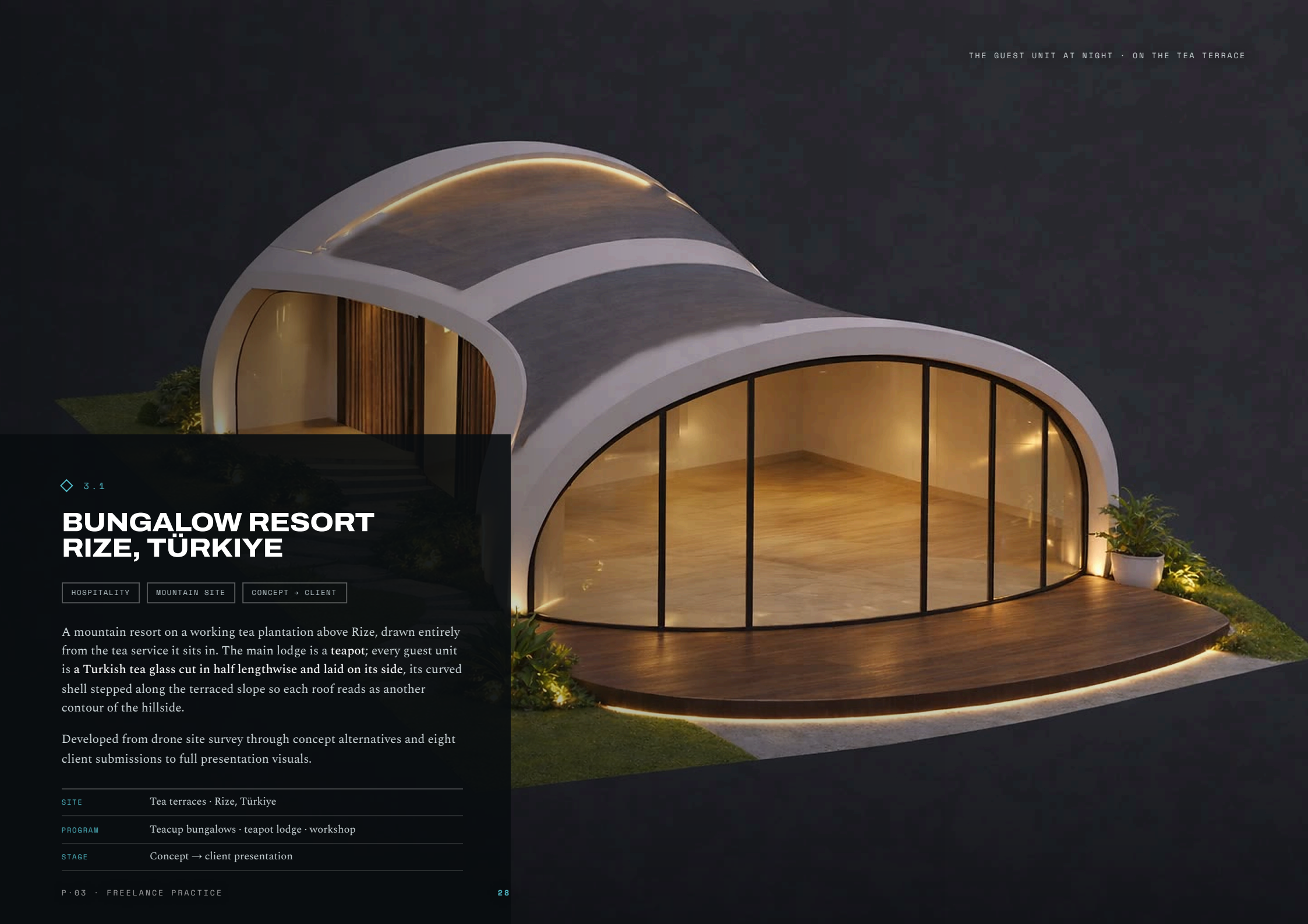
Design and visualisation commissions carried alone from brief to final image — each with its own client, its own site and its own character.

CLIENTS
DESIGN-BUILD · PRIVATE

ROLE
DESIGN · VISUALISATION

STAGE
CONCEPT → CLIENT PRESENTATION

PLACES
RIZE · TRIPOLI · BEIRUT



THE GUEST UNIT AT NIGHT · ON THE TEA TERRACE

◇ 3.1

BUNGALOW RESORT RIZE, TÜRKIYE

HOSPITALITY | MOUNTAIN SITE | CONCEPT → CLIENT

A mountain resort on a working tea plantation above Rize, drawn entirely from the tea service it sits in. The main lodge is a **teapot**; every guest unit is **a Turkish tea glass cut in half lengthwise and laid on its side**, its curved shell stepped along the terraced slope so each roof reads as another contour of the hillside.

Developed from drone site survey through concept alternatives and eight client submissions to full presentation visuals.

SITE	Tea terraces · Rize, Türkiye
PROGRAM	Teacup bungalows · teapot lodge · workshop
STAGE	Concept → client presentation

3.1

THE HILLSIDE, TWICE

The plot was flown before it was drawn. The drone survey set the terrace lines, and the units were plotted straight onto the frame that recorded them.

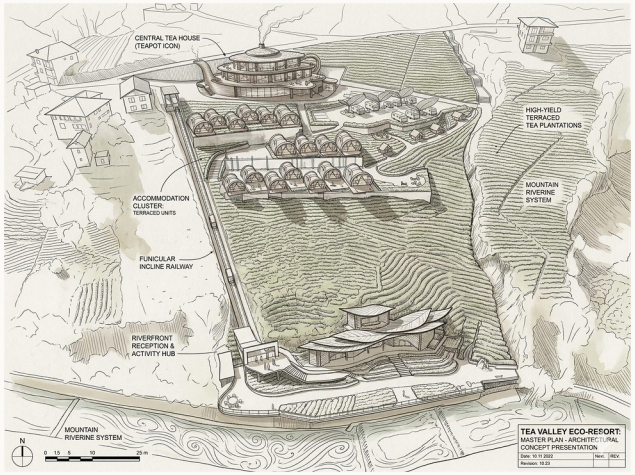
What the ground gave, the scheme kept: no terrace was cut flat to suit a building. The units sit along the contour lines the tea already follows, which is why the roofs read as another row of the plantation rather than as objects dropped on it.

The hand studies came first and stayed useful — the site strategy and the schematic master plan were what the client actually argued with, across eight submissions.

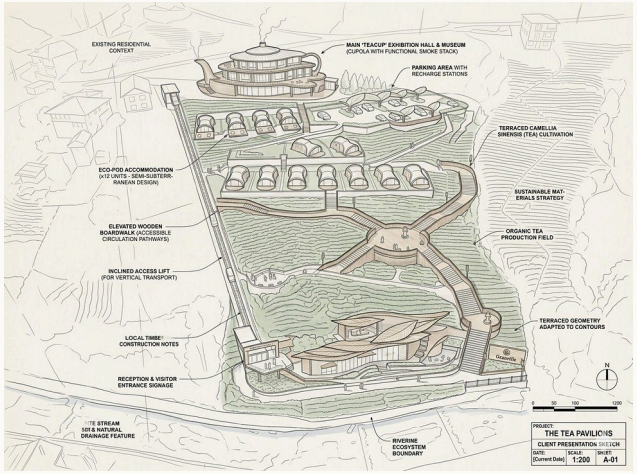
SURVEY	Drone photogrammetry
METHOD	Units plotted on the real frame
SUBMISSIONS	Eight, to the client

THE SITE AS FOUND

UNITS PLOTTED ONTO IT



MASTER PLAN · HAND STUDY



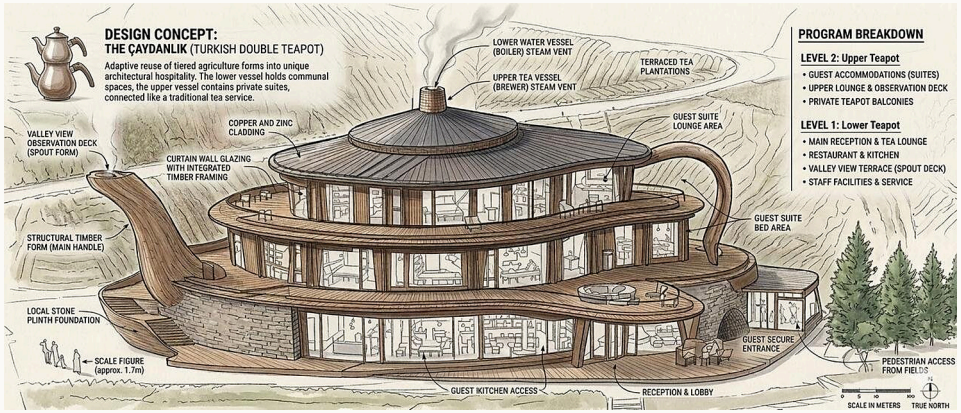
THE SCHEMATIC SKETCH

3.1

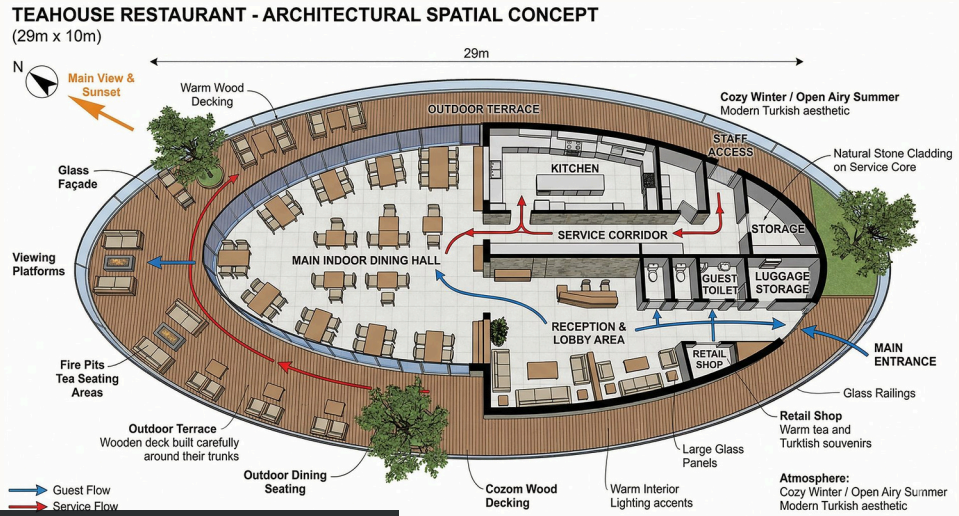
TEAPOT AND TEACUP

The concept is literal and it is drawn that way. The lodge is a teapot, cut open floor by floor; the guest unit is a tea glass halved lengthwise and laid down, its curved shell carrying a green roof into the slope.

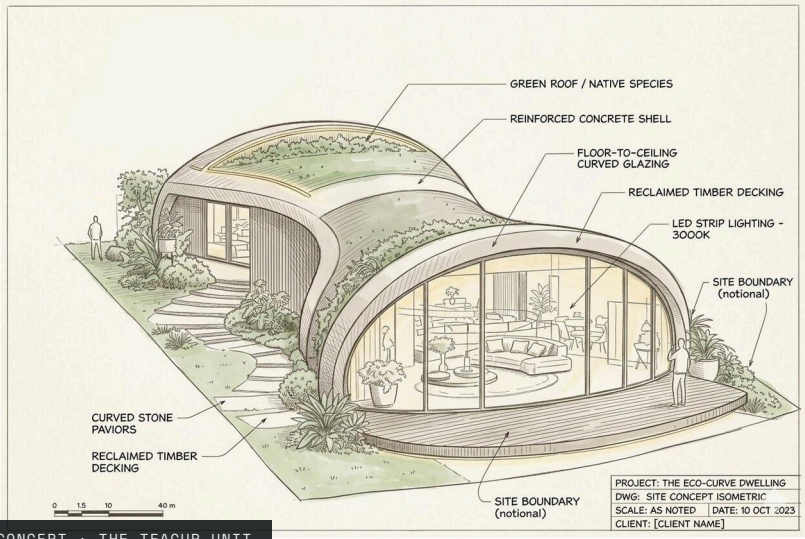
THE TEAPOT LODGE · SECTIONAL CUTAWAY



DESIGN CONCEPT · THE TEAPOT LODGE



TEAHOUSE RESTAURANT · SPATIAL CONCEPT



DESIGN CONCEPT · THE TEACUP UNIT

3.1

THE UNIT ITSELF

Cut a Turkish tea glass in half down its length, lay it on its side, and you have the unit: a curved shell that is widest at the window wall and narrows to the back of the plan.

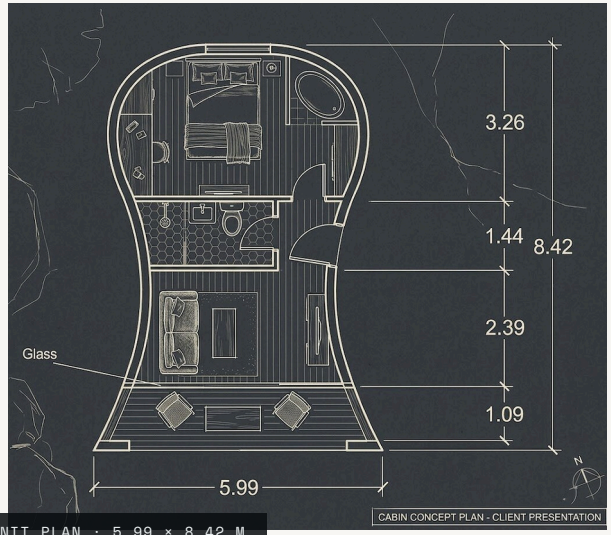
The plan keeps the glass silhouette too — the waisted **hourglass** outline of an *ince belli* glass, drawn out at 5.99 × 8.42 m for the compact unit and 6.47 × 10.03 m for the family one.

The shell is reinforced concrete under a green roof, so the unit carries the plantation over itself rather than interrupting it.

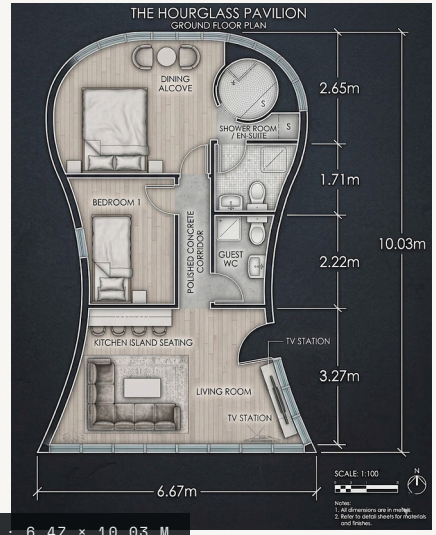
UNITS	Compact + family
ROOF	Green roof over curved shell
STRUCTURE	Reinforced concrete shell

THE UNIT FORM · A HALF TEA GLASS ON ITS SIDE

THE SHELL, FROM THE BACK



UNIT PLAN · 5.99 × 8.42 M



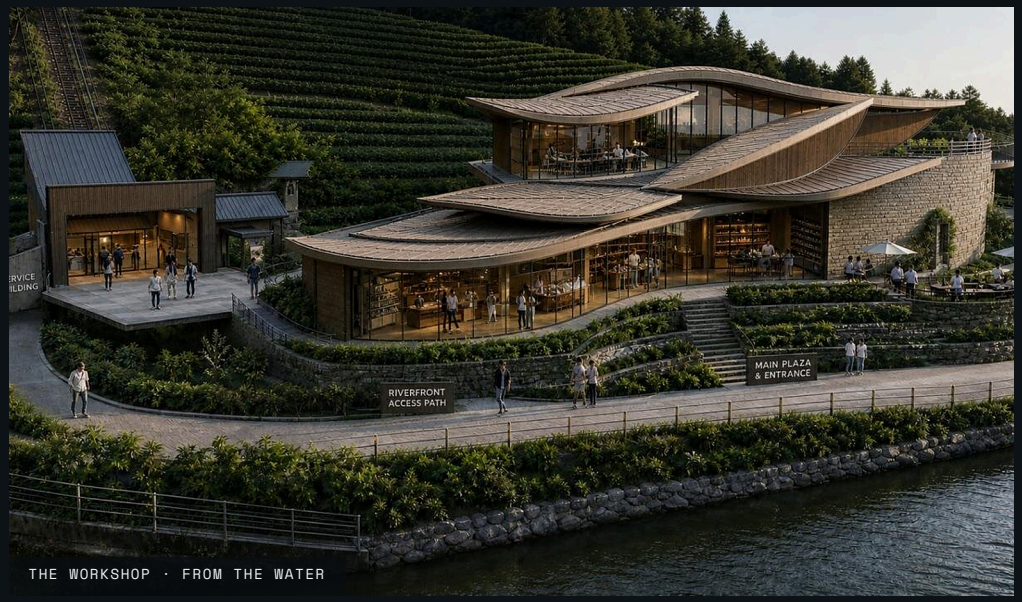
UNIT PLAN · 6.47 × 10.03 M

3.1

THE WORKSHOP AND THE TERRACES

The tea workshop sits lowest, on the river terrace, where the leaf comes off the slope. Above it the lodge and the units step up the plantation.

TEA WORKSHOP · ON THE RIVER TERRACE



THE WORKSHOP · FROM THE WATER

THE RESORT ON THE TERRACES · AERIAL

THE VALLEY · PERSPECTIVE STUDY

3.2 – 3.4

CINEMA, LIBRARY, INTERIORS

Three smaller commissions, each carried alone from brief to final image. A cinema drum whose glowing crown does the marketing; a winter library in two structural characters, a cantilevered entry against a pitched reading hall; and two living-room interiors.

CINEMA · THE NIGHT CROWN

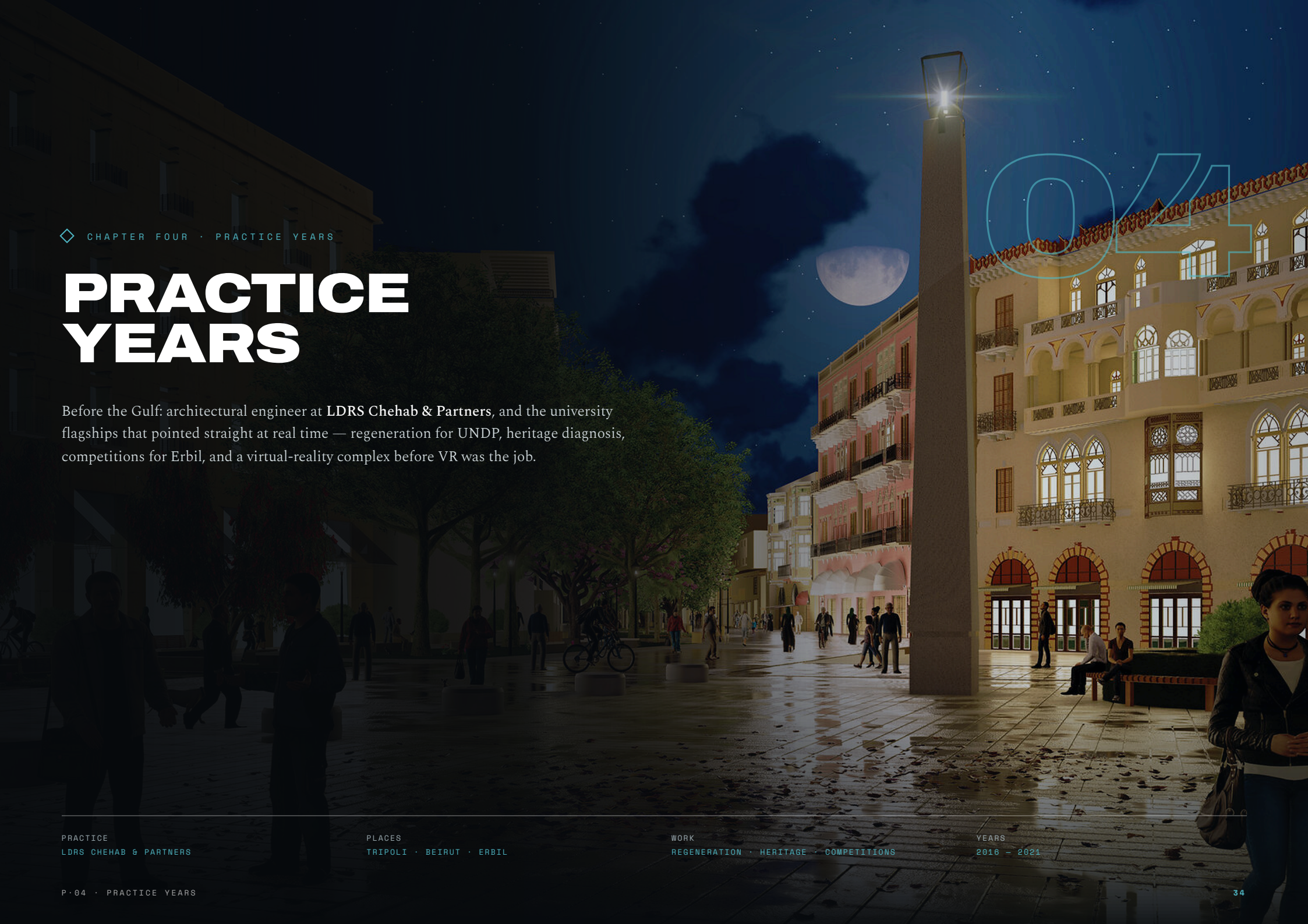
CINEMA · FROM THE BOULEVARD

LIBRARY · THE CANTILEVER ENTRY

LIBRARY · THE READING HALL

LIVING · EUROPEAN

LIVING · MINIMAL



CHAPTER FOUR · PRACTICE YEARS

PRACTICE YEARS

Before the Gulf: architectural engineer at **LDRS Chehab & Partners**, and the university flagships that pointed straight at real time — regeneration for UNDP, heritage diagnosis, competitions for Erbil, and a virtual-reality complex before VR was the job.

PRACTICE	PLACES	WORK	YEARS
LDRS CHEHAB & PARTNERS	TRIPOLI · BEIRUT · ERBIL	REGENERATION · HERITAGE · COMPETITIONS	2016 — 2021

4.1

SKYFALL PAVILION

COMPETITION | VVIP PALACE | PERFORATED CANOPY

A VVIP palace beneath a single perforated canopy — **"a waterfall of sky"** — filtering Kurdistan's light into a palm court. Sixteen boards carried the philosophy, the sun studies, the terraced gardens and the plans; the court render is the argument in one image.

CONCEPT	شلال السماء · a waterfall of sky
PROGRAM	Palace · gardens · reception

THE PAVILION · WIDE PANORAMA

ACROSS THE LAWN

NIGHT AERIAL

ACROSS THE LAKE · DUSK

4.2

ROYAL COMPLEX ERBIL

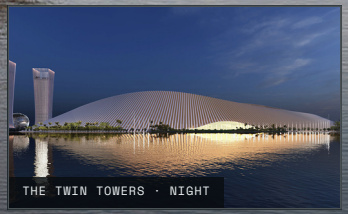
COMPETITION | 44-PAGE BOOK | RETAIL + TOWERS

A lakeside “eyelid” mall crowned by a cable-stayed canopy, paired with **twin perforated towers holding a suspended sphere** — delivered as a full 44-page competition book: masterplan, programme acts, structure studies and the renders.

CLIENT	Iraq Head of Ministry
DELIVERABLE	Competition book · visuals

ATRIUM · UNDER THE CANOPY

THE ROOF GARDEN


THE TWIN TOWERS · NIGHT

4.3

AL TALL REGENERATION

3DS MAX | CORONA | LADYBUG / HONEYBEE | PHOTOGRAMMETRY

A UNDP-backed regeneration of the Ottoman clock-tower square at the heart of Tripoli. I authored the **full 3D reconstruction and every visualisation**, then ran the science underneath: solar comfort, wind, shadow and pedestrian simulations that decided where people would actually sit, walk and stay.

CLIENT	UNDP · Municipality of Tripoli
PRACTICE	LDRS Architects
ROLE	Full 3D · visualisation · analysis

4.3

THE SQUARE AT EYE LEVEL

The scheme was delivered as a full walkthrough film — the square the way a Tuesday afternoon would actually feel. These are frames from it, and from the render set beside it.

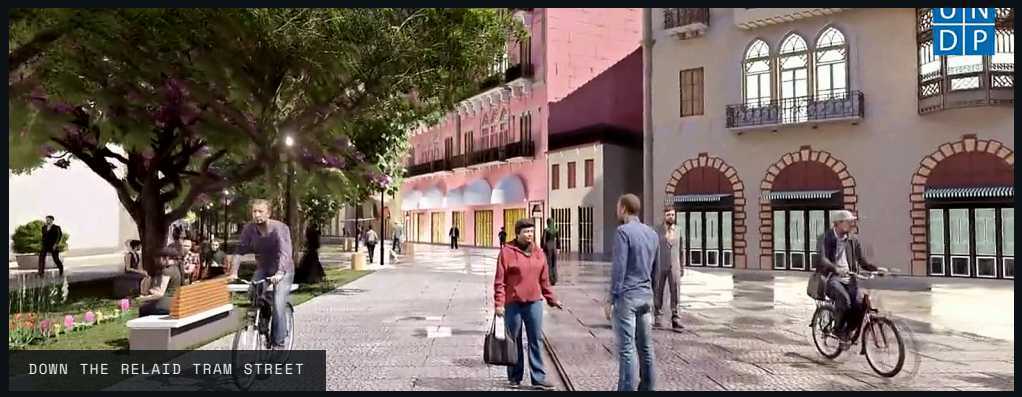
THE FOUNTAIN COLONNADE

THE HERITAGE FRONTAGE

THE TERRACED GARDENS

THE PROMENADE

THE CLOCK TOWER FROM THE WATER GARDEN



DOWN THE RELAID TRAM STREET

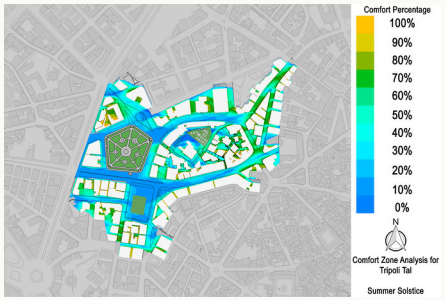
THE WHOLE DISTRICT, REBUILT

TRIPOLI · THE OTTOMAN CORE RECONSTRUCTED FROM PHOTOGRAMMETRY, THEN REGENERATED

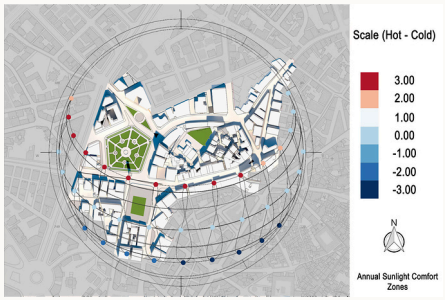
4.3

THE EVIDENCE LAYER

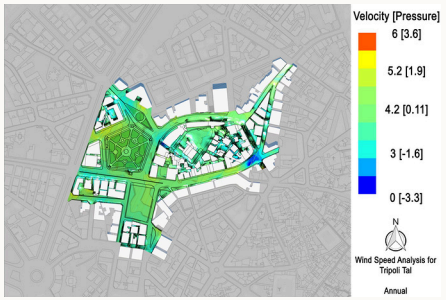
Design decisions argued with data — comfort zones, solstice shadows, wind velocity, pedestrian stay-places, commercial heat, and **algorithmically placed café locations** scored against the comfort map. Some layers were walked and counted by hand; others simulated in Grasshopper.



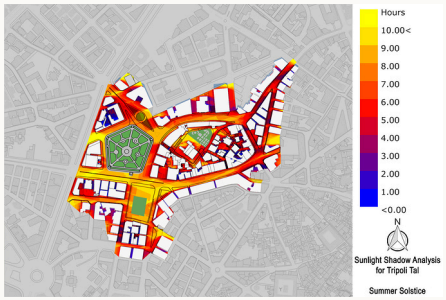
COMFORT ZONES · SUMMER



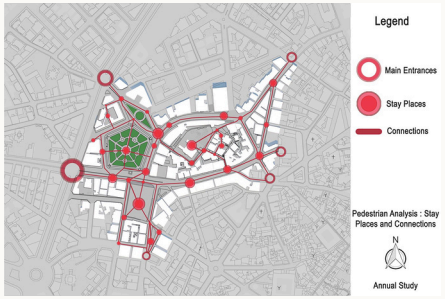
ANNUAL COMFORT



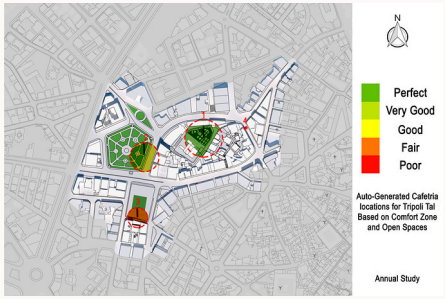
WIND VELOCITY



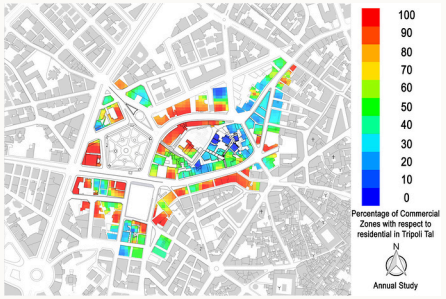
SHADOW HOURS · SOLSTICE



PEDESTRIAN STAY-PLACES



AUTO-PLACED CAFÉS



COMMERCIAL HEATMAP

PHOTOGRAMMETRY BASE · THE REAL DISTRICT

Comfort. Annual and solstice comfort maps set where a person can actually stand still in July. The planting, the colonnade and the water were placed against those maps, not against a plan drawing.

Wind and shade. Velocity and pressure across the square, and shadow-hour counts at the solstices, decided the depth of the arcade and the height of the tree canopy.

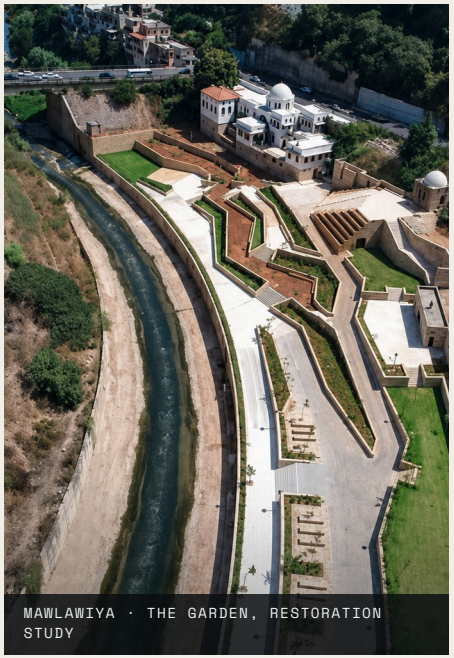
People and trade. Stay-places were walked, counted and mapped by hand; the commercial-to-residential heatmap came from the same field notes. Café positions were then generated against the comfort surface rather than chosen.

HERITAGE: TWO RECORDS

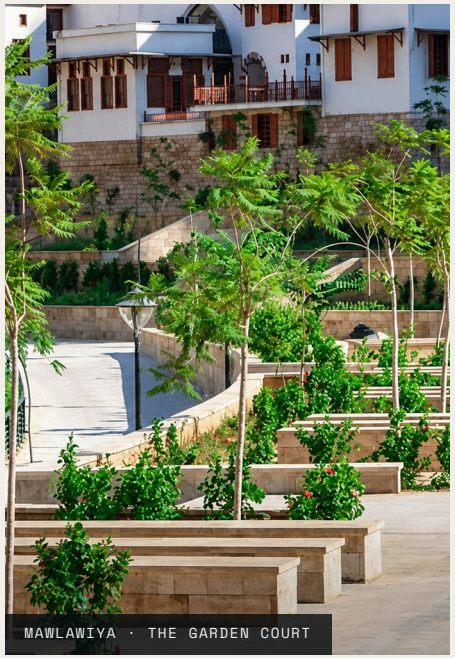
Al Mawlawiya and the old souks. Rehabilitation of the Mevlevi lodge gardens and the old souk edge in Tripoli — stone terraces, water channels and planting brought back around the historic tekke. Worked from the drawings to the ground: detailed design, bills of quantities, and supervision of the works on site.

Mar Semaan Monastery, 1878. A field diagnosis in Zghorta — walking the vaults and walls to document cracks, settlement, water infiltration and fire damage, then mapping the pathologies and recommending anchoring and tie-rod interventions to hold the stone together.

PRACTICE	LDRS Chehab & Partners
SCOPE	Drawings · BOQ · supervision
METHOD	On-foot survey · photo record



MAWLAWIYA · THE GARDEN, RESTORATION STUDY



MAWLAWIYA · THE GARDEN COURT

MAWLAWIYA · TERRACES, STUDY VIEW

MAR SEMAAN · THE MONASTERY

MASONRY RECORD

INTERIOR SURVEY

4.6

VIRTUAL REALITY COMPLEX

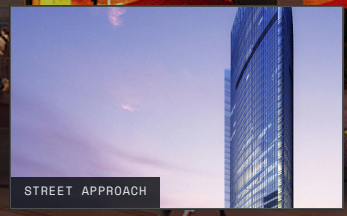
ARCH540 | MIXED USE | DATA-DRIVEN | 97 PAGES

A VR entertainment and education complex — **the first of its scale proposed in the region** — argued with a sixty-person survey, demographic infographics and programme analysis before a line was drawn, then delivered as a full urban set-piece with towers, an arena ring and a public realm.

COURSE	ARCH540 · Senior Project
METHOD	Survey · data · then design
EXTRAS	Animation · research booklet

THE ARENA BOWL

THE COMPLEX ON THE SHORE


STREET APPROACH

4.6

ARGUED ON PAPER FIRST

Ninety-seven pages of survey, demographics and programme analysis sat under the design before a line was drawn.

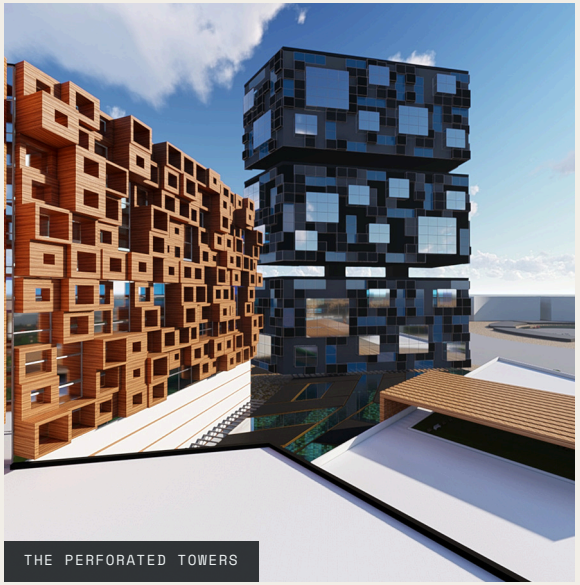
A sixty-person survey set the programme: what people in the region would actually come to a virtual-reality complex to do, how often, and at what price. The demographic infographics decided the mix; the mix decided the massing.

The whole argument — research, programme, plans, structure and the renders — was then delivered as a single presentation board, with the research carried alongside it as a bound booklet.

COURSE	ARCH540 · Senior Project
RESEARCH	97 pages · 60-person survey
DELIVERED	One board · one booklet

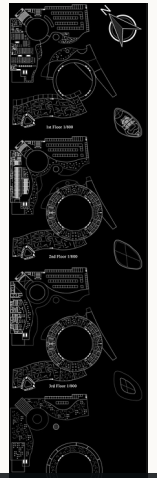
THE FULL PRESENTATION BOARD

THE RESEARCH BOOKLET



THE PERFORATED TOWERS

THE BOULEVARD · DUSK



PLANS · BASEMENT → FOURTH FLOOR

TWO UNIVERSITY FLAGSHIPS

Antalya skyscraper — the cliff edge. Antalya's cliffs, missing their marker. The tower fuses a sky tower and a sea tower into one stratified monolith grown out of the rock face — anchored horizontally back into the cliff, and drawn to 1:800 below the waterline in sea elevation and sea section.

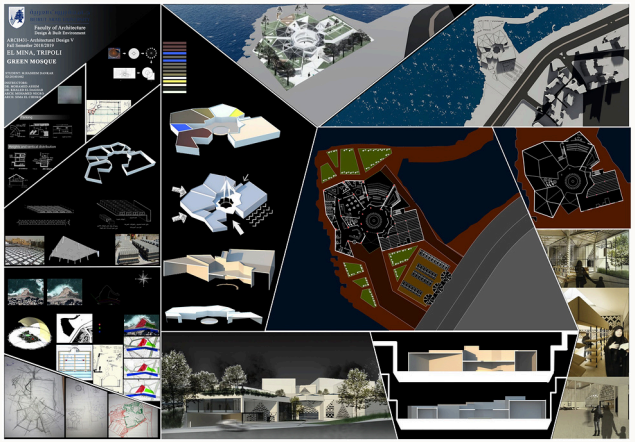
Green Mosque. A mosque folded into a green star on the El Mina corniche — prayer hall, amphitheatre plaza and landscape terraces resolved in one faceted ground between the road and the sea.

ANTALYA	ARCH432 · Design VI
MOSQUE	ARCH431 · Design V
SITES	Düden cliffs · El Mina, Tripoli

ANTALYA · IN SITE, THE DÜDEN CLIFFS

GREEN MOSQUE · THE STAR ON THE CORNICHE

ANTALYA · THE TWO PERSPECTIVES



GREEN MOSQUE · THE FULL BOARD

4.9

ALGORITM ROME

Two weeks at Sapienza University of Rome: forty hours of Algorithmic-Aided Design under **Arturo Tedeschi**, then forty hours designing and physically fabricating a smart kinetic façade — the first contact with the computational tools that now anchor the national twin.

THE STUDIO

FABRICATING THE FAÇADE

PROTOTYPE ASSEMBLY

THE FINAL JURY

4.10

THE PAPER TRAIL

Verifiable, dated, signed — the documents behind the curriculum vitae.

RIBA

Royal Institute of British Architects

Mohamad Hashem Jamal Dankar
5th Floor, Dankar Building, Infront of Zahrat Al Qalamoun Market
Main Sea Road
Qalamoun
Tripoli
LEBANON

4th November 2020

To Whom It May Concern:

Statement of RIBA Recognised Architectural Qualification

This statement is issued at the request of **Mohamad Hashem Jamal Dankar** who is listed in RIBA's records as having been awarded the following qualification, which is validated by the Royal Institute of British Architects:

- Bachelor Degree of Architectural Engineering, validated at RIBA Part 1 level, awarded by Beirut Arab University (BAU), Tripoli Campus, Lebanon, in July 2020.

Beirut Arab University (BAU) is one of over eighty schools worldwide whose qualifications are validated against RIBA's qualification framework.

Please do not hesitate to contact me should you require any further information.

Yours faithfully

Caroline Gooch
RIBA Education Department
education@riba.org

Royal Institute of British Architects
66 Portland Place,
London, W1B 1AD, UK

Tel: +44 (0)20 7580 5533
education@riba.org
www.architecture.com

RIBA · PART 1 STATEMENT

Certificate شهادة

The Faculty of Architecture - Design & Built Environment certifies that
Mohamed Hashem Jamal Dankar ID 201601402
born in Kalamoun; April 4th 1998
has successfully completed the requirements,
and received The Degree of Bachelor of Architecture
in the spring of 2019 / 2020 with a CGPA 2.62

Beirut, 21-07-2020

BAU · B.ARCH CERTIFICATE

FiBREX
Construction Group since 1972

02nd November 2023

Experience Certificate

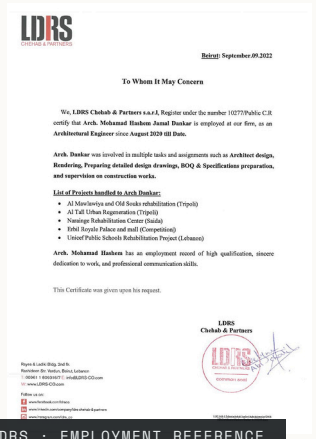
This is to certify that Mr. Mohamad Hashem Dankar holder of Lebanese Passport No LR1830558 was employed in our company as Architect from 14th February 2023 to 26th October 2023.

This certification is being issued upon his request for any purpose it may serve him.

Yours faithfully,
for Fibrex Contracting

Sajmer Abi Darghami
Executive Director

FIBREX · EXPERIENCE CERTIFICATE


LDRS
CHEHAB & PARTNERS

Beirut: September.09.2022

To Whom It May Concern

We, LDRS Chehab & Partners s.a.r.l, Register under the number 10277/Public C.R certify that Arch. Mohamad Hashem Jamal Dankar is employed at our firm, as an Architectural Engineer since August 2020 till Date.

Arch. Dankar was involved in multiple tasks and assignments such as Architect design, Rendering, Preparing detailed design drawings, BOQ & Specifications preparation, and supervision on construction works.

List of Projects handled by Arch Dankar:

- Al Mawlawiya and Old Souks rehabilitation (Tripoli)
- Al Tall Urban Regeneration (Tripoli)
- Nanajjar Rehabilitation Center (Saida)
- Ethel Boyajis Palace and mall (Competition)
- Unicef Public Schools Rehabilitation Project (Lebanon)

Arch. Mohamad Hashem has an employment record of high qualification, sincere dedication to work, and professional communication skills.

This Certificate was given upon his request.

LDRS
Chehab & Partners

LDRS · EMPLOYMENT REFERENCE

Algoritm

Algoritm Rome Program

This is to certify that:

MOHAMAD HACHEM DANKAR

has successfully completed the Algoritm Rome Summer School Program
From 17th of July to 29th of July 2017
of 80 Total Training Hours

Algoritm Organizer · SAPIENZA Host · Authenticator

Arturo Tedeschi · Mostafa Rabea

SAPIENZA ROME · PARAMETRIC PROGRAMME

AUTODESK CERTIFICATION

AUTODESK.

In recognition of a commitment to professional excellence

Mohamad Hashem Dankar

has successfully completed the Autodesk Certified requirements of:

Autodesk Certified User: 3ds Max®

AUTODESK. · verify.certiport.com · January 29, 2019

AUTODESK · 3DS MAX

AUTODESK CERTIFICATION

AUTODESK.

In recognition of a commitment to professional excellence

Mohamad Hashem Dankar

has successfully completed the Autodesk Certified requirements of:

Autodesk Certified User: AutoCAD®

AUTODESK. · verify.certiport.com · January 29, 2019

AUTODESK · AUTOCAD

IC3
DIGITAL LITERACY CERTIFICATION

CERTIPORT

MOHAMAD HASHEM DANKAR

Key Applications · Living Online · Computing Fundamentals

JULY 24, 2019
DATE ISSUED

verify.certiport.com

IC3 GS5 · DIGITAL LITERACY

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LinkedIn Learning
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Sunnyvale, CA 94085

RHINO 7 · COURSE

شهادة مشاركة

تفيد جمعية صفدي الثقافية

بان الشاب محمد هاشم جمال دنكر

قد شارك في ورشات عمل تصميم
وصنع المقاعد والأحواض الخاصة بمشروع

Our City Our Way

العامل بتمويل مشترك من الاتحاد الأوروبي في إطار مشروع
SouthMed CV

OUR CITY OUR WAY · DESIGN-BUILD

◆ CONTACT

LET'S BUILD IT *twice.*

EMAIL

hashemx98@gmail.com

PHONE / WHATSAPP

+971 50 970 2011

BASE

Abu Dhabi

United Arab Emirates

PORTFOLIO

The live build runs

the same work in 3D

THE LIVE PORTFOLIO RUNS THIS WORK IN THE BROWSER — WALKTHROUGH FILM, A SCROLL-DRIVEN PHOTOGRAMMETRY CAPTURE, AND REAL ENGINE ASSETS YOU CAN ORBIT IN 3D.

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